

Entropy Signatures of Market Stress: What Survives a Calibrated Null

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Abstract

Entropy-based crisis indicators jump in crises and decay slowly afterwards, and this sawtooth is often read as the trace of information that arrives in bursts and is absorbed gradually. We show that most of this signature is a measurement artefact. We decompose the entropy of a return vector into three channels — scale, dependence ($\frac{1}{2} \log \det R$) and shape (the entropy deficit $\mathcal{D} = D_{KL}(p \parallel q)$) — and track the scale-free structural index $\mathcal{J} = \mathcal{D} - \frac{1}{2} \log \det R = D_{KL}(p \parallel \prod_i q_i) \geq 0$ on twenty S&P 500 stocks (1990–2022) and on Kenneth French’s 48 industry portfolios (1990–2026). Every statistic is judged against two calibrated nulls, two hundred replications each, that resample the observed returns: one i.i.d., one in 21-day blocks. Jump asymmetry, the skewness signature, channel coupling, the volatility–dependence offset and the regime gap of $\mathcal{J}$ itself are all reproduced, sometimes exceeded, by resampled data: a positive convex functional of estimated moments jumps up as soon as a fat-tailed observation enters the estimator. A single result clears both nulls on both panels: under stress, the cross-section couples. The regime gap of the dependence channel is -0.99 nat against -0.21 and -0.51 under the nulls (stocks), and -3.65 against -0.56 and -2.46 (industries). The relaxation half-life is not identified, and no channel improves out-of-sample risk forecasts. Finally we present a Kullback–Leibler-ball stress-testing construction whose radius is calibrated on the market’s realised information flow; it prices classical scenarios in nats, but its severity scale is milder than history, so its return periods are not validated.

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Notation

Two warnings before the table. First, differential entropy h **depends on units**: $h(aX) = h(X) + \log|a|$, so expressing returns in percent rather than in decimals shifts h by a constant. Divergences ($D_{KL}(\cdot \| \cdot)$, I , $\mathcal{D}$, $\mathcal{J}$) are on the contrary invariant under any invertible change of variable: that is the reason the structural index exists. Second, K denotes a kurtosis and the letters “KL” are the initials of Kullback and Leibler; the two are unrelated.

Symbol	Meaning	Nature
p_t	true conditional distribution of returns at t	—
q_t	Gaussian with the same mean and covariance as p_t	—
$h(\cdot)$	differential entropy	nats, <i>unit-dependent</i>
$D_{KL}(p \ q)$	Kullback–Leibler divergence	nats, invariant
$\mathcal{D}_t$	entropy deficit = $D_{KL}(p_t \ q_t) \geq 0$	nats, invariant
$\mathcal{D}_1(z)$	deficit of a standardised scalar (negentropy)	nats, invariant
H_t^{vol}	scale channel = $\sum_i \log \sigma_{i,t}$	nats, <i>unit-dependent</i>
H_t^{dep}	dependence channel = $\frac{1}{2} \log \det R_t \leq 0$	nats, invariant
H_t^{cov}	$H_t^{vol} + H_t^{dep}$; the constant $\frac{N}{2} \log 2\pi e$ is omitted	nats
$\mathcal{J}_t$	structural index = $\mathcal{D}_t - H_t^{dep} \geq 0$	nats, invariant
$I(X; Y)$	mutual information	nats, invariant
$I_t^{(h)}$	information flow = $D_{KL}(p_t \ p_{t-h})$	nats
S, K	skewness, excess kurtosis	dimensionless
$\Sigma_t = D_t R_t D_t$	covariance, standard deviations, correlation	—
$\lambda_k, N_t^{\text{eff}}$	eigenvalues of R_t , effective number of modes	$1 \leq N^{\text{eff}} \leq N$
$\kappa, \sigma_{\mathcal{J}}$	relaxation rate and diffusion coefficient of $\mathcal{J}$	—
η	radius of a stress scenario	nats
N	number of assets	—

1 Introduction

Two facts about financial prices sit uneasily together. The first is that the Gaussian model is the natural answer to a question of ignorance: fix a mean and a variance, know nothing else, and the maximum-entropy distribution is Gaussian. The second is that returns are not Gaussian: they have fat tails, negative skewness, and their dependence tightens precisely when it matters.

The usual reaction is to treat the second fact as a defect of the first model and to patch it: add jumps to the price, add a stochastic volatility factor, add a copula. This paper goes the other way. If the Gaussian is what maximum entropy delivers, then any departure from the Gaussian measures the market’s distance from maximal ignorance, that is, it measures information. The question is then not how to patch the Gaussian but how to read the size, the dynamics and the direction of the departure.

The paper is organised around an asymmetry in that departure. Information arrives in bursts and is absorbed gradually. A central-bank surprise, a default, a war removes uncertainty in an instant; the market then spends weeks working out the implications, and uncertainty rebuilds. If entropy is the right currency of uncertainty, its dynamics should be a sawtooth: sharp drops, slow recoveries. That shape is not a stylised fact that has to be assumed — half of it is a theorem. Conditioning cannot increase expected entropy, so the arrival of a signal lowers the *average* conditional entropy by exactly the mutual information between signal and price (Proposition 2); and the maximum-entropy ceiling weakly falls at every realisation, without the fall being quantifiable in general (Proposition 3). The two statements are distinct and Appendix E separates them explicitly,

with a worked Gaussian example. The *expected* size of the jump, in nats, is the information content of the news. What has to be postulated is the recovery and the direction.

Three contributions follow.

The volume bridge (Section 2). The step from “maximum entropy gives a Gaussian” to “prices are Gaussian” is often asserted and rarely argued, because maximum entropy is a statement about a distribution at fixed variance and says nothing about where that variance comes from. We close it by defining the price as accumulated order flow and applying the principle in volume time: total traded volume, which is observable, fixes the variance budget, and the maximum-entropy increment conditional on volume is Gaussian with variance proportional to volume. This is the mixture-of-distributions hypothesis (Clark, 1973; Tauchen & Pitts, 1983; Ané & Geman, 2000), derived rather than assumed. Temporal consistency then makes the log-price a Brownian motion subordinated to the volume clock, and Itô’s lemma makes the price a geometric Brownian motion. The bridge also explains, in the same sentence, why calendar-time returns are not Gaussian: either the conditional distribution is not maximum-entropy (information constrains), or the volume clock is random, or both — and the two sources leave different fingerprints.

Three channels and a scale-free index (Section 3). The Gaussian entropy of a return vector splits exactly into a scale term and a dependence term, and the true entropy differs from it by the entropy deficit $\mathcal{D} = D_{KL}(p \parallel q)$, where skewness and fat tails live. The scale term dominates the level of differential entropy and says nothing about information; removing it leaves $\mathcal{J} = \mathcal{D} - \frac{1}{2} \log \det R$, which is exactly $D_{KL}(p \parallel \prod_i q_i)$ — the divergence from a product of Gaussian marginals with the same scales. $\mathcal{J}$ is positive, invariant to the level of volatility, and additive over its two entropy-destroying channels. It is the state variable of the rest of the paper. Empirically it shares only 21% of its variance with volatility, and its shape channel is essentially orthogonal to volatility ($r = 0.09$).

Stress tests in nats (Section 8). A scenario of severity η is the worst conditional distribution inside a KL ball of radius η around today’s. The construction is not new (Glasserman & Xu, 2014; Breuer & Csizsár, 2013); the calibration is. The radius is read off the market’s realised information flow, the h -step revision $D_{KL}(p_t \parallel p_{t-h})$, measured in the same unit as the radius, so that a severity acquires a return period in principle; Section 8.2 shows that on our data this return period is not validated. Because the mean shift and the variance inflation are chosen from one budget, the scenario cannot be internally inconsistent, and because *any* scenario can be priced on the same scale, the question “is this stress test severe enough?” becomes arithmetic.

An exploratory extension (Appendix F). The KL ball fixes the severity of the *portfolio* and is silent on how the underlying covariance should deform, asset by asset, to reach it. An eighth hypothesis, which clears only the i.i.d. null, is used to calibrate that split on an exploratory basis; we put it in an appendix so that it does not weigh on the main conclusions.

A warning about measuring any of this. The structural index is a positive convex functional of estimated second, third and fourth moments. Estimation noise on fat-tailed data pushes such a functional *up* sharply and lets it decay *smoothly*, which is exactly the sawtooth the postulates predict. Every jump and relaxation statistic below is therefore reported against a null built by resampling the observed returns — keeping the marginal tails, destroying the chronology. The result is uncomfortable and, we think, the most useful thing in the paper: the jump asymmetry, which looks overwhelming, does not survive, nor does the relaxation half-life; the coupling of the

cross-section under stress and the amplitude of $\mathcal{J}$ do. Anyone building a crisis indicator out of a convex functional of estimated moments should run this null before believing their own chart.

1.1 Position in the literature

What is not new. None of the mathematical ingredients is, and we prefer to say so up front. Deriving price dynamics from an entropic inference principle, rather than assuming them, is the programme of “entropic dynamics”, applied to stocks and options (Abedi & Bartolomeo, 2019) and then to jump-diffusion processes (Abedi, 2026); the volume bridge of Section 2 only adds the volume clock of Clark (1973) and Ané & Geman (2000). The decomposition of Section 3 is the classical identity of independent component analysis (Comon, 1994): the divergence from a product of Gaussian marginals splits into marginal negentropies and multi-information, and $-\frac{1}{2} \log \det R$ is the total correlation of Watanabe (1960) of a Gaussian vector. “Conditioning reduces entropy” is a textbook fact (Cover & Thomas, 2006). We claim only the assembly of these pieces into a scale-free index and the testing discipline that follows.

Crisis indicators built on the correlation structure. The only result that survives our nulls — the cross-section couples under stress — is the best-established stylised fact of the crisis literature: correlations rise when markets fall (Longin & Solnik, 2001). It is measured by the variance share of the leading eigenmode (the “absorption ratio” of Kritzman et al., 2011), by principal-component and Granger-causality connectedness measures (Billio et al., 2012), by spectral or centrality entropies of the correlation matrix (Almog & Shmueli, 2019; Chakraborti et al., 2019; Kukreti et al., 2020), by a complexity gap from random matrix theory (Mukhia, Ansari & Nurujjaman, 2026), by entropy-based early-warning indicators (Billio et al., 2016), and, in central-bank practice, by composite stress indices (Hollo, Kremer & Lo Duca, 2012). Our dependence channel $\frac{1}{2} \log \det R$ is a member of this family, expressed in nats; we do not claim that it dominates the absorption ratio, and the head-to-head comparison remains to be done. It also inherits a known problem: under heteroskedasticity, correlation measured on a high-volatility subsample is mechanically higher, even at constant underlying correlation (Forbes & Rigobon, 2002). Section 7.2 says how far our nulls address it.

Entropy and crises, without a null. A large literature reports that measures of entropy, mutual information or divergence move during crises (Zhou, Cai & Tong, 2013; Caraiani, 2018; Papla & Siedlecki, 2024; Noguer i Alonso, 2025). To our knowledge these studies compare the measure to zero or to a calm period, rarely to a resampling of the same returns. Surrogate-data tests are nevertheless the norm in nonlinear time-series analysis (Theiler et al., 1992; Schreiber & Schmitz, 2000). The main contribution of this paper is to apply that discipline to an entropy-based crisis indicator and to publish the result, which is largely negative.

Entropic stress tests. Searching for the worst scenario inside a Kullback–Leibler ball is the construction of Breuer & Csiszár (2013), close to the plausibility measure of Breuer et al. (2009), to the minimum-relative-entropy updating of Meucci (2008) and to the robust risk measure of Glasserman & Xu (2014); Glasserman, Kang & Kang (2015) select scenarios by empirical likelihood. What we add is the calibration of the radius on the market’s realised information flow, and the pricing in nats of existing scenarios. Section 8.2 shows that this calibration is too mild against history.

Outline. Section 4 states the theorem and the two postulates and derives the jump-diffusion dynamics; Section 5 lists the hypotheses; Sections 6–7 give the data, the estimators, the nulls and the results; Section 8 builds the stress test; Section 9 is candid about what none of this establishes. Appendix F gathers the eighth hypothesis and the calibrated composition of scenarios, as exploratory material.

2 From order flow to a Gaussian price: the volume bridge

2.1 The price as accumulated order flow

Over $[t, t + \Delta]$, let M be the number of executed trades, trade k carrying signed volume $\epsilon_k v_k$ with $\epsilon_k \in \{-1, +1\}$ its sign (buyer- or seller-initiated) and $v_k > 0$ its size. Write $V = \sum_k v_k$ for total volume and $Q = \sum_k \epsilon_k v_k$ for net signed order flow.

Assumption 1 (Linear impact). *The log-price responds linearly to net signed order flow, $\Delta X_{t,\Delta} = X_{t+\Delta} - X_t = \lambda Q$, with $\lambda > 0$ constant over the interval.*

This is the equilibrium of Kyle (1985): the market maker, unable to tell informed from uninformed flow, sets a linear pricing rule and λ is the inverse of depth. It is the definition of “price as a function of traded volume” that the rest of the section needs, and it is the weakest link of the chain; Section 2.5 says what happens when it breaks.

Assumption 2 (Exchangeable signs and variance proportional to volume). *Conditional on V and on the information available at t , the market’s belief about Q is constrained only by its first two moments, with $\mathbb{E}[Q \mid V] = m$ and $\text{Var}(Q \mid V) = \zeta^2 V$.*

The form of the variance is what one obtains when trade signs are exchangeable with bounded correlation and sizes have a stable mean: for independent signs, $\text{Var}(Q) = \sum_k v_k^2 \approx \bar{v}V$. The economic content is that *volume, not time, is the natural measure of what can happen*.

2.2 Maximum entropy in volume time

Proposition 1 (Maximum entropy under first- and second-moment constraints). *Among absolutely continuous distributions on $\mathbb{R}^N$ with mean μ and covariance $\Sigma \succ 0$, the Gaussian $\mathcal{N}(\mu, \Sigma)$ uniquely maximises differential entropy, and $h^{\max}(\Sigma) = \frac{1}{2} \log((2\pi e)^N \det \Sigma)$.*

Proof. Let $q = \mathcal{N}(\mu, \Sigma)$ and let p have the same first two moments and finite differential entropy. Since $\log q$ is exactly quadratic, its expectation depends only on those moments, so $\mathbb{E}_p[\log q] = \mathbb{E}_q[\log q] = -h(q)$. Positivity of relative entropy gives $0 \leq D_{KL}(p \parallel q) = -h(p) + h(q)$, hence $h(p) \leq h(q)$, with equality if and only if $p = q$ almost everywhere. The argument requires neither differentiability of p nor the prior existence of a maximiser. $\square$

The variational route — a Lagrangian on the normalisation, mean and covariance constraints, whose first-order condition makes $\log p$ quadratic — explains *why* the maximiser is Gaussian, but it is the divergence argument that establishes optimality and uniqueness.

Applying Proposition 1 to Assumption 2 in one dimension, then transporting through Assumption 1:

$$\Delta X_{t,\Delta} \mid V \sim \mathcal{N}(\lambda m, \lambda^2 \zeta^2 V). \quad (1)$$

Equation (1) is the bridge. Three points deserve emphasis.

First, *the conditioning variable is volume, not time*. Maximum entropy does not deliver a variance; it delivers the least committal distribution given a variance. Assumption 2 supplies that variance from an observable, and that observable is traded volume. Equation (1) is the mixture-of-distributions hypothesis of Clark (1973) and Tauchen & Pitts (1983), obtained here as a maximum-entropy statement rather than posited; Ané & Geman (2000) show empirically that returns measured in transaction time are close to Gaussian while calendar-time returns are not, which is the prediction of (1).

Second, *Gaussianity is conditional*. Unconditionally, ΔX is a scale mixture of normals over the random volume V , hence leptokurtic even if (1) holds exactly. This is the subordination mechanism of Mandelbrot & Taylor (1967), and it means that fat tails alone are no evidence against maximum entropy.

Third, the multivariate extension is immediate: with N assets, $\Delta \mathbf{X} \mid V \sim \mathcal{N}(\mu V, \Sigma V)$, and Proposition 1 applies as is.

2.3 From Gaussian increments to Brownian motion and to GBM

Let $\theta_t = \int_0^t v_s ds$ be the cumulative volume clock. Equation (1) gives the law of one increment; a process requires more.

Assumption 3 (Temporal consistency). *In the volume clock, increments over disjoint intervals are independent, paths are continuous, and the covariance of an increment is proportional to elapsed volume.*

Under Assumption 3 with constant coefficients, there is a matrix B with $BB^\top = \Sigma$ and $X_\theta = X_0 + \mu\theta + BW_\theta$ in volume time, hence in calendar time

$$X_t = X_0 + \mu\theta_t + BW_{\theta_t}, \quad (2)$$

a Brownian motion *subordinated* to the volume clock. If trading intensity is constant, $\theta_t = \bar{v}t$ and (2) is an ordinary Brownian motion in calendar time; exponentiating and applying Itô,

$$\frac{dS_{i,t}}{S_{i,t}} = (a_{i,t} + \tfrac{1}{2}\sigma_{i,t}^2) dt + \sigma_{i,t} dW_{i,t}, \quad (3)$$

so that the drift of the price is not that of the log-price — the same second-order correction that reappears in Section 3 when $\log \det R_t$ is differentiated.

It is worth being precise about what Assumption 3 does. By the theorem of Monroe (1978), *every* continuous local martingale is a time-changed Brownian motion: subordination alone therefore has no empirical content. The content is in the clock: (1) says the right clock is volume, and Assumption 3 says the market is memoryless in that clock. It is maximum entropy that selects Brownian motion within the class that Monroe's theorem allows.

2.4 The contrapositive

Proposition 1 reads both ways. Let q_t be the Gaussian with the same conditional mean and covariance as p_t . Then

$$\mathcal{D}_t \equiv h(q_t) - h(p_t) = D_{KL}(p_t \parallel q_t) \geq 0, \quad (4)$$

with equality if and only if p_t is Gaussian. Hence: *prices are non-Gaussian exactly to the extent that their conditional distribution is not maximum-entropy*, and $\mathcal{D}_t$ measures that extent. In the

language of Section 2, a positive $\mathcal{D}_t$ means that constraints beyond the first two moments are active — that someone knows something the second moment does not encode.

Two sources produce a positive deficit, and separating them matters. A random volume clock produces a scale mixture, symmetric and leptokurtic. Binding directional information produces skewness. Section 3 splits $\mathcal{D}$ exactly along that line.

2.5 Where the bridge is weak

Assumption 1 is linear; empirical metaorder impact is closer to a square root of size, which makes the map from flow to price concave and the induced return distribution less fat-tailed than the flow. Assumption 2 treats signs as exchangeable; order flow is in fact strongly autocorrelated, which inflates $\text{Var}(Q)$ above $\varsigma^2 V$ and makes the volume clock an imperfect proxy for variance. None of these failures changes the *form* of the argument — maximum entropy under a variance budget remains Gaussian — but both mean that the volume clock is an approximation to the information clock rather than that clock itself. We use volume-time reasoning to motivate the framework and calendar-time estimation in all the empirical work, so that no result below depends on the exactness of (1).

3 The three channels of entropy

3.1 Scale and dependence

Write $\Sigma_t = D_t R_t D_t$ with $D_t = \text{diag}(\sigma_{1,t}, \dots, \sigma_{N,t})$. Then $\det \Sigma_t = \left(\prod_i \sigma_{i,t}^2\right) \det R_t$ and

$$h(q_t) = \underbrace{\frac{N}{2} \log(2\pi e)}_{\text{constant}} + \underbrace{\sum_i \log \sigma_{i,t}}_{H_t^{\text{vol}}} + \underbrace{\frac{1}{2} \log \det R_t}_{H_t^{\text{dep}}}. \quad (5)$$

By Hadamard's inequality, $H_t^{\text{dep}} \leq 0$, with equality if and only if $R_t = I$: *correlation always destroys entropy*, and $-H_t^{\text{dep}}$ is precisely the Gaussian multi-information (total correlation) of the return vector. A useful reading of the same quantity is spectral: with λ_k the eigenvalues of R_t and $\tilde{\lambda}_k = \lambda_k/N$, the participation number $N_t^{\text{eff}} = \exp\left(-\sum_k \tilde{\lambda}_k \log \tilde{\lambda}_k\right)$ is the effective number of independent risk modes; it equals N for $R = I$ and tends to 1 when a single collective mode absorbs the variance.

Equation (5) is the ceiling, not the entropy. It is what the covariance matrix alone implies, and Proposition 1 says the truth lies below it.

3.2 Shape: the entropy deficit and its composition

Subtracting (4) from (5) gives the paper's decomposition, three channels and a constant:

$$h(p_t) = \frac{N}{2} \log(2\pi e) + \underbrace{H_t^{\text{vol}}}_{\text{scale}} + \underbrace{H_t^{\text{dep}}}_{\text{dependence}} - \underbrace{\mathcal{D}_t}_{\text{shape}}. \quad (6)$$

The shape channel is skewness and fat tails, and the Edgeworth expansion says so explicitly: for a standardised variable with skewness S and excess kurtosis K near Gaussianity,

$$J \approx \frac{S^2}{12} + \frac{K^2}{48}. \quad (7)$$

Equation (7) answers the question “where would an entropy jump be visible?”: in the third and fourth moments, additively, with a skewness channel and a tail channel. It is also the source of the sign problem that Section 4 must solve, since S enters squared — *negentropy cannot tell a crash from a rally*. As an estimator, (7) is unusable on daily returns; Section 6.2 documents how badly and what we use instead.

3.3 The structural index

The scale channel is a nuisance for our purpose: it dominates the level of (6) numerically, it moves with volatility, and a change of units alters it. The other two channels are scale-free. Combining them, and using the exact chain rule $D_{KL}(p \parallel \prod_i q_i) = \sum_i D_{KL}(p_i \parallel q_i) + I(x)$ where I is the multi-information:

Definition 1 (Structural information index).

$$\mathcal{J}_t \equiv \mathcal{D}_t - H_t^{dep} = D_{KL}(p_t \parallel \prod_i q_{i,t}) \geq 0, \quad (8)$$

the divergence of the conditional joint law of returns from a product of Gaussian marginals with the same scales.

$\mathcal{J}$ is zero if and only if returns are independent and Gaussian; it rises when assets couple, when marginals fatten or when they skew. It is measured in nats, it is invariant to the level of volatility, and it is additive over its two channels, so that one can always ask how much of a move comes from dependence and how much from shape. It is the state variable of Section 4.

4 Information dynamics: one theorem, two postulates

4.1 Downward jumps are a theorem

Two distinct statements underlie the downward half of the postulated dynamics. Conflating them is an easy mistake, and earlier versions of this work made it; we therefore separate them explicitly.

Recall first the definition of the *mutual information* between the return vector X and a signal Y :

$$I(X; Y) = D_{KL}(p_{XY} \parallel p_X \otimes p_Y) = \iint p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dx dy, \quad (9)$$

that is, the divergence of the joint law from the product of the marginals: how far the joint law is from independence. It is positive, symmetric, zero if and only if $X \perp Y$, and — unlike h — invariant under invertible reparametrisation. For a scalar Gaussian pair with correlation ρ , $I = -\frac{1}{2} \log(1 - \rho^2)$.

Proposition 2 (Fall in average conditional entropy). *Let X be the return vector and Y a signal. Then*

$$\mathbb{E}_Y[h(X \mid Y)] = h(X) - I(X; Y) \leq h(X),$$

with equality if and only if $X \perp Y$.

Proposition 3 (Monotonicity of the ceiling under constraint). *Let $\mathcal{C} \subseteq \mathcal{C}'$ be two sets of constraints and $\mathcal{Q}(\cdot)$ the set of laws satisfying them. Then $\mathcal{Q}(\mathcal{C}') \subseteq \mathcal{Q}(\mathcal{C})$ and therefore*

$$\sup_{p \in \mathcal{Q}(\mathcal{C}')} h(p) \leq \sup_{p \in \mathcal{Q}(\mathcal{C})} h(p).$$

Both results are classical (Cover & Thomas, 2006) and they are not equivalent; the detailed proof and a worked Gaussian example are in Appendix E. Their scopes differ along two axes.

Proposition 2 *quantifies* the fall — exactly $I(X; Y)$ nats — but only *on average* over realisations of the signal. For a particular y , $h(X | Y = y)$ can exceed $h(X)$: a sufficiently confusing signal leaves the market more uncertain than before. One should therefore expect a jump component that is mainly, not exclusively, one-sided — which is what Section 7 finds.

Proposition 3 holds on the contrary *for every realisation*, but concerns the maximum-entropy *ceiling* rather than the actual entropy, and it quantifies nothing: it says the ceiling cannot rise, not by how much it falls.

What the two say together, and what motivates everything that follows: the fall is *discontinuous*, because the signal arrives at a point in time, and its expected size in nats is the information content of the news — an amplitude measured on a scale that depends neither on the asset, nor on the currency, nor on units.

A caveat in fairness. Neither proposition is used to compute anything in the empirical part. We measure $\mathcal{J}$, not $h(X) - \mathbb{E}_Y[h(X | Y)]$: estimating a differential entropy in dimension 20 on a 252-day window is out of reach (Appendix A). The link between the theorem and the statistics reported below is an argument, not an estimated identity, and the reader should take it as such.

4.2 Two postulates

Postulate 1 (Relaxation). In the absence of new information, the structural index decays towards a long-run level at a constant rate: $d\mathcal{J}_t = -\kappa (\mathcal{J}_t - \bar{\mathcal{J}}) dt + \sigma_{\mathcal{J}} dW_t$ with $\kappa > 0$.

The economics behind P1 is that a constraint wears off. The news is priced in, disagreement reforms, positions disperse, and the cross-section recovers the freedom the shock took away. Nothing in information theory delivers this; it is the paper’s substantive hypothesis, and κ is the object to estimate.

Why an Ornstein–Uhlenbeck process and not something else? The choice is not one of convenience: it is *forced* once three minimal requirements are imposed, and it is the only one that satisfies all three.

1. *Markov.* The future of $\mathcal{J}$ depends on the past only through its present level. This is the parsimony requirement: postulate no memory beyond what the current state already contains.
2. *Stationary.* $\mathcal{J}$ has a finite long-run level and variance. Without this the index drifts, it is not a state variable, and the regime comparisons of Section 7 are meaningless.
3. *Gaussian noise.* Between two arrivals of information, we claim to know nothing about the noise beyond its variance — the maximum-entropy principle applied to the noise itself, exactly as it was applied to returns in Section 2.

The theorem of Doob (1942) states that **the Ornstein–Uhlenbeck process is the only process that is at once Markov, Gaussian and stationary** in continuous time. The three requirements therefore leave no latitude for the diffusive part: it is determined. The jump term, for its part, is added precisely to *break* Gaussianity where the framework says it fails, namely at arrivals of information. The structure of (11) is thus an imposed skeleton plus a deliberately non-Gaussian component, which is exactly the class introduced by Barndorff-Nielsen & Shephard (2001).

What, on the other hand, remains arbitrary. Three things, which should be stated rather than left to look as if they followed from the theorem. The rate κ is assumed *constant*: nothing forbids a crisis from being absorbed faster, or slower, than ordinary news, and the data used here cannot decide. The pull is *linear* in the gap to the long-run level, whereas an accelerated pull far from the mean would be just as plausible. Finally, $\mathcal{J}$ is positive by construction whereas an Ornstein–Uhlenbeck with a Brownian term can become negative: the specification of Barndorff-Nielsen & Shephard (2001) avoids this by dropping the Brownian term and driving the process with a positive-jump subordinator, which guarantees a stationary law supported on $\mathbb{R}^+$. Empirically the question does not arise — $\mathcal{J}$ stays between 2.17 and 21.13 nats on the main panel and between 0.42 and 1.69 on the auxiliary European panel, very far from the boundary — but formally one should either drop $\sigma_{\mathcal{J}}$ or write the process on $\log \mathcal{J}$.

Finally, the postulate is not innocuous from a measurement standpoint: an estimator with memory will manufacture decay even in the absence of any economic relaxation, so Section 6.3 measures the decay that a memoryless world would already produce.

Postulate 2 (Asymmetry of the constraint). The constraint that information imposes is one-sided in the direction of losses, so that an entropy jump comes with negative skewness of the market factor.

Why this postulate is logically necessary. The index $\mathcal{J}$ is an *even* function of skewness: flipping the sign of skewness does not change its value. The reason is immediate. The shape channel is $\mathcal{D}_1(z) = D_{KL}(p_z \parallel \varphi)$, and the normal law φ is symmetric; so if z is replaced by $-z$, the reference law is unchanged and so is the divergence:

$$\mathcal{D}_1(-z) = \mathcal{D}_1(z), \quad \text{and therefore} \quad \mathcal{J}(-x) = \mathcal{J}(x). \quad (10)$$

The Edgeworth expansion (7) makes this visible to the eye: skewness S enters squared.

The consequence is structural, not anecdotal. **A distribution with a long left tail — a crash — and its mirror image with a long right tail — a rally — have exactly the same $\mathcal{J}$.** The index measures the *size* of information, never its direction. No refinement of the estimator will change this: it is a property of the object measured. If the paper wants to say anything about direction, it must therefore add an outside ingredient. P2 is that ingredient, and it is the smallest possible one: a single claim, about the sign, added to a framework that by construction produces none.

What it rests on. Not a mechanism we would assert, but an established empirical regularity: the *leverage effect* in the sense of Black (1976), namely that negative returns are followed by larger volatility increases than positive returns of the same size. It is one of the most robust stylised facts of empirical finance (Christie, 1982; Engle & Ng, 1993; Nelson, 1991). In the language of this paper: for the same amount of information, the market’s response is larger on the downside than on the upside, so a jump in $\mathcal{J}$ arrives together with negative market skewness.

We deliberately abstain from choosing between competing causal explanations of that effect — financial leverage proper, volatility feedback, or asymmetric withdrawal of liquidity on the sell side. The postulate depends on none of them: it asserts only the asymmetry, not its cause. It is testable separately (H5) and refutable separately, and Section 7 indeed concludes that it is not established.

4.3 The resulting dynamics

Proposition 2 together with P1 gives a non-Gaussian Ornstein–Uhlenbeck process with one-sided jumps, of the class introduced by Barndorff-Nielsen & Shephard (2001) for stochastic volatility:

$$d\mathcal{J}_t = -\kappa(\mathcal{J}_t - \bar{\mathcal{J}})dt + \sigma_{\mathcal{J}}dW_t + \int_{z>0} z\mathcal{P}(dt, dz), \quad (11)$$

The integral term needs careful reading, as its notation is compact.

What $\mathcal{P}$ is not. It is neither a Gaussian law nor a function. The letter $\mathcal{N}$ is reserved in this paper for the normal law, so the jump measure carries a distinct symbol.

What $\mathcal{P}$ is. A *random counting measure*, known as a Poisson random measure. One can picture it as a random scatter of points in the (date, size) plane: each point is an information event, its abscissa the date at which it occurs and its ordinate the size of the jump it causes. Formally, $\mathcal{P}(dt, dz)$ equals 1 if there is a jump in the time interval $[t, t + dt)$ whose size falls in $[z, z + dz)$, and 0 otherwise. The integral $\int_{z>0} z\mathcal{P}(dt, dz)$ then reads simply: “if there is a jump at date t , add its size z to $\mathcal{J}$; otherwise add nothing”. It is a compact way of writing $\sum_j z_j$, the sum of the jumps that occurred.

What characterises it. Two ingredients, combined in its *intensity measure* $\nu(dz)dt$, the expected number of jumps of size between z and $z + dz$ during dt : an *arrival rate* $\lambda = \nu(\mathbb{R}^+)$, in jumps per unit time, and a *size law* $\nu(dz)/\lambda$. On the main panel, Section 7 estimates $\lambda \approx 5$ jumps a year, a mean size of 0.62 nat and a maximum of 3.21 nats.

What z is. The size of an individual jump, in nats. Conceptually, it is the mutual information of the news that caused it, in the sense of Proposition 2. Operationally, it is the change in $\mathcal{J}$ picked up by the four-sigma filter of Section 6.2. The two coincide only approximately, and the paper does not claim to identify them.

Why the integral runs only over $z > 0$. Because $\mathcal{J}$ measures an entropy *deficit* and, by Proposition 2, information can only increase that deficit in expectation. In this framework nothing creates a *sudden increase* in entropy: entropy is restored only by the slow relaxation of the first term, never by a jump. Hence one-sided jumps.

Two caveats, however. Proposition 2 is a statement *on average*: a particularly confusing realisation can increase entropy, so a faithful specification would allow a small probability of negative jumps rather than forbid them. And the Poisson measure assumes independent arrivals, whereas the event study of Section 7 shows clear clustering of jumps; a self-exciting Hawkes-type process would be more faithful.

Note finally that the diffusion coefficient is written $\sigma_{\mathcal{J}}$ and not η , the latter letter being reserved for the radius of stress scenarios (Section 8). Read on entropy itself rather than on $\mathcal{J}$, the sign flips: entropy falls by jumps and recovers by diffusion. The observable predictions are sharp, and they are what Section 5 tests: the law of $\Delta\mathcal{J}$ is right-skewed; large moves are up and small moves down; and after an upward jump the index decays with half-life $\ln 2/\kappa$.

4.4 Why fat tails, correlation and jumps arrive together

The framework yields a prediction that no volatility or correlation model yields on its own. It follows in four steps, without technical vocabulary.

1. Important news concerns *all* assets at once: the same information constrains each of them.
2. If the news is bad, all assets therefore fall *on the same day*.
3. That day also appears, taken in isolation, in the history of each asset: for it, it is an unusually large loss.
4. Accumulating such days deforms each asset’s return distribution: the loss side lengthens and thickens more than the gain side.

Step 2 is what the dependence channel measures: assets move together, measured correlation rises, $-H^{dep}$ rises. Step 4 is what the shape channel measures: more and larger extreme losses than extreme gains, that is, higher kurtosis and negative skewness, so $\mathcal{D}$ rises. The two channels rise together because they describe the same day seen from two places — the cross-section for one, each asset’s distribution for the other.

In calm markets, notable moves are specific to each asset: they do not align the cross-section and so have no reason to raise both channels at the same time.

The testable form is a *regime-dependent coupling*: the correlation between $-\Delta H^{dep}$ and $\Delta \mathcal{D}$ should be close to zero in calm markets and strongly positive under stress. That is H6, and it is where the paper answers the question “does the link between fat tails, correlation and entropy jumps require an extra postulate?”: no. Given Proposition 2, the coupling follows from the common nature of the news. Only the *sign* requires P2.

5 Hypotheses

H1 (regime signature). In the stress regime, the dependence channel falls ($H^{dep} \downarrow$), the shape channel rises ($\mathcal{D} \uparrow$) and hence the structural index rises ($\mathcal{J} \uparrow$).

H2 (partial offset). Changes in the two channels of the Gaussian ceiling are *opposite in sign and proportional*: there is a coefficient β such that

$$\Delta H_t^{dep} = -\beta \Delta H_t^{vol} + u_t, \quad \mathbb{E}[u_t] = 0, \quad (12)$$

and the hypothesis concerns the *sign*: $\beta > 0$. The coefficient β is the *offset rate*: the fraction of a scale-entropy move absorbed by dependence. It reads directly — $\beta = 0$ means no offset, $\beta = 1$ full offset and hence conservation of $\det \Sigma$, $0 < \beta < 1$ partial offset, $\beta < 0$ amplification. A secondary hypothesis concerns regimes: $\beta_{\text{stress}} > \beta_{\text{calm}}$.

This formulation replaces that of earlier versions of this project, which asserted in one block that “ H^{cov} is more stable than its components”. That statement mixed two distinct claims — a sign and a magnitude — and could only be tested in a binary way. Separating them makes it possible to *report* a coefficient with its confidence interval rather than pronounce a verdict, and makes the comparison across datasets quantitative rather than contradictory.

H3 (jump asymmetry). Changes in $\mathcal{J}$ are one-sided: at a common threshold, upward jumps are more numerous and heavier than downward jumps, and $\Delta \mathcal{J}$ is right-skewed — *beyond what the same estimator produces on data with the same marginal law and no temporal structure*. The qualifier is the whole hypothesis; without it the statement is untestable.

- H4 (relaxation).** $\mathcal{J}$ mean-reverts with $\kappa > 0$ and a half-life appreciably longer than the memory of the estimator itself, judged against the same null.
- H5 (skewness signature).** Upward jumps in $\mathcal{J}$ coincide with negative market skewness and with a rise in the skewness channel (postulate P2).
- H6 (channel coupling).** The dependence and shape channels are weakly coupled in calm markets and strongly coupled under stress.
- H7 (predictive power).** Knowing today’s entropy says something about the risk of the next 21 days that today’s volatility alone does not already say. “Improves” is meant exclusively on dates the model had never seen at estimation time; Section 7.8 spells out the protocol.
- H8 (composition of the crisis budget, exploratory).** Grouped into episodes, jumps in $\mathcal{J}$ have the *dependence* channel as their dominant contributor more often than a resampling without temporal structure produces. H8 is judged against the i.i.d. null only and serves only to calibrate the composition of a scenario; it is treated in Appendix F.

6 Data and estimation

6.1 Datasets

The main panel gathers twenty S&P 500 constituents at daily frequency, from 3 January 1990 to 28 December 2022 (8,312 return days; 8,061 rolling dates after a 252-day window), from the `sp500` dataset distributed with the `skfolio` package. These are *individual stocks*, not industry portfolios. The choice is deliberate: aggregating firms into industries mechanically imposes common-factor dependence, and an earlier version of this project cited a single-stock analysis as the robustness check it most needed. The period covers the Asian and LTCM episodes of 1997–98, the technology crash, the global financial crisis, the euro-area debt crisis, 2015–16, 2018, COVID-19 and the 2022 correction.

The second panel is Kenneth French’s daily 49-industry file (`config/default.yaml`; the residual category is excluded, hence 48 portfolios), 1990–2026, about 9,000 dates. The full pipeline was run on it, including both nulls at two hundred replications; the tables are in `paper/tables/ff49_*`. Contrasting individual stocks with industry portfolios is the useful contrast: a signal that holds on both is not merely an aggregation effect. Only these two panels carry verdicts against the nulls.

An auxiliary panel, `EuStockMarkets` (Bollerslev & Ghysels, 1996), serves as illustration: daily closing levels of the DAX, SMI, CAC and FTSE, 1,860 trading days from 1991 to 1998, $N = 4$. It is far too small to carry a result, has not been run against the two-hundred-replication nulls, and after the burn-in period of the stress threshold only 1,104 dates remain, 160 of them in stress. We report its numbers when they shed light on a point, never as a replication.

6.2 Estimators

Covariance. Two estimators run in parallel. The reference is a 252-day rolling window with Ledoit–Wolf shrinkage (Ledoit & Wolf, 2004), which reproduces the level statistics of the earlier version. The responsive estimator, used for everything to do with jumps, is exponentially weighted with a 60-day half-life plus 10% shrinkage towards the diagonal. The choice matters: with a rectangular window, an extreme observation *leaving* the sample produces a discontinuity in the entropy series on a date when nothing happened, which would contaminate any jump test. Exponential weights remove this artefact by construction.

Negentropy. The shape channel is estimated marginally, asset by asset, with the bounded-contrast approximation of Hyvärinen (1998), whose two terms split naturally into an odd (skewness) channel and an even (tail weight) channel. The Edgeworth form (7) is kept for interpretation only, because it fails badly as an estimator exactly where financial data live. Table 1 makes that failure explicit against a nonparametric reference.

Law	S	K	J (Vasicek)	J Edgeworth	J Hyvärinen
Gaussian	−0.00	0.03	0.012	0.000	0.000
Uniform	0.00	−1.20	0.179	0.030	0.065
Skew normal ($a = -4$)	−0.80	0.67	0.074	0.062	0.049
Laplace	−0.01	2.95	0.087	0.181	0.087
Student $t(8)$	−0.02	1.63	0.030	0.056	0.016
Student $t(5)$	0.10	5.13	0.061	0.549	0.052
Student $t(4)$	−0.00	18.34	0.102	7.006	0.105
Lognormal ($s = 0.5$)	1.76	5.78	0.203	0.954	0.167
Exponential	2.00	6.11	0.424	1.112	0.268

Table 1: Marginal negentropy in nats, $n = 200,000$ draws. The reference is the Vasicek spacing estimator of differential entropy. The Edgeworth form $S^2/12 + K^2/48$ overstates a Student- $t(4)$ by a factor of 69; the Hyvärinen approximation is within 3% of the truth there. Daily equity returns sit squarely in the $t(4)$ – $t(5)$ region, which is why every deficit reported below uses the latter. Reproduced by `scripts/negentropy_benchmark.py`.

Two limitations deserve to be stated rather than buried. Estimating marginal deficits and then adding the linear dependence term captures marginal non-Gaussianity and Gaussian-copula dependence, but omits nonlinear tail dependence; we measure it separately through an average pairwise co-exceedance rate. And the Hyvärinen approximation, accurate in the leptokurtic region, understates the negentropy of a moderately fat $t(8)$ by about half, so the levels reported below are conservative.

Jump detection. A jump in $\mathcal{J}$ is flagged when $|\Delta\mathcal{J}_t|$ exceeds four times a rolling local scale built on bipower variation (Barndorff-Nielsen & Shephard, 2004), which is consistent for the diffusive scale even in the presence of jumps and is lagged by one day so that the threshold judging a date never contains it.

Regimes without look-ahead. The stress regime is the top decile of rolling 21-day market volatility. The threshold at date t is the 90% quantile of that volatility over dates strictly *before* t (expanding quantile), and the first 504 dates, lacking history, belong to no regime. An earlier version of this project used the full-sample quantile, which uses future volatility to label the past and inflates regime gaps (Section 7.1). The expanding threshold classifies more dates as stress (1,076 out of 7,557 instead of 807) because market volatility tends to rise over the sample: “stress” means “more turbulent than 90% of everything seen so far”, which is more honest and changes the definition. The same rule applies to the data and to the nulls.

Inference. A 252-day rolling window means two consecutive observations share 251 of their 252 underlying return days. Every regime comparison therefore uses a 252-day circular block bootstrap, under a null that breaks the value–label link while preserving the persistence of the series and the clustering of the regime indicator. Predictive comparisons use expanding-window walk-forward estimation refitted annually, evaluation on non-overlapping dates, and a Diebold–Mariano test (Diebold

& Mariano, 1995) on the squared-error differential. Three conventions of this protocol were corrected relative to the earlier version: the p -value of the relaxation slope is that of the Dickey–Fuller (MacKinnon) distribution, because under a unit root the normal law over-rejects; HAC lags follow the Newey–West rule ($\lfloor 4(n/100)^{2/9} \rfloor$, i.e. 10 here) except for forecast targets that overlap over 21 days (H7), where they are at least 21; and the day after a jump is no longer excluded from the relaxation estimate, because excluding it selected on the dependent variable; the measured effect is negligible (120.1 then 120.8 days on the main panel).

6.3 The placebo null

The structural index is a positive convex functional of estimated second, third and fourth moments. Estimation noise alone pushes it *up* sharply and lets it decay *smoothly* — which is exactly the signature that (11) predicts. A jump-asymmetry statistic or a relaxation half-life is therefore evidence only relative to this mechanical null, never relative to zero.

We measure this null in two ways. First a controlled simulation (`scripts/estimator_memory.py`): i.i.d. Gaussian returns with constant covariance — a world without any dynamics — passed through the production pipeline. A single injected stress day produces a 2.13-nat spike in $\mathcal{J}$ that decays with a half-life of **35 trading days**, purely through estimator memory; on clean data without any event, the four-sigma detector flags **37 upward jumps and zero downward jumps** and $\Delta\mathcal{J}$ has a skewness of +2.9. The mechanical asymmetry is real and it is large.

Second, two calibrated nulls (`scripts/placebo_null.py`) that resample the rows of the observed return matrix, preserving the cross-sectional dependence and fat marginal tails of the real data. The i.i.d. version destroys everything the postulates are about — volatility clustering, time-varying dependence, information-arrival dynamics. The block version resamples 21-day blocks, so that short-term clustering survives and only longer-horizon regime dynamics are destroyed. Two hundred replications of each give the null distributions reported in Table 2. Both apply the identical volatility filter to define stress regimes, so that the regime comparisons of H1, H2 and H6 are tested on the same footing as they are estimated.

7 Results

7.1 What the calibrated null leaves standing

Table 2 is the paper’s central empirical result and it is uncomfortable. Two nulls are built, both by resampling the observed return matrix and running the identical pipeline — including the volatility filter that defines the stress regime — two hundred times each, on each of the two panels.

The i.i.d. null resamples rows independently. It keeps fat marginal tails and cross-sectional dependence and destroys *all* temporal structure. It asks: beyond the fact that turbulent windows contain large returns, does the measure detect anything?

The block null resamples circular 21-day blocks. It additionally keeps short-term volatility clustering and destroys only longer-horizon regime dynamics. It asks the harder question: beyond clustering, does the measure detect information dynamics?

Neither null is nested in the other in a way that would make one uniformly conservative, and they disagree often enough that reporting only one would be misleading. A statistic that clears both is evidence; a statistic that clears one and fails the other is not identified.

Statistic	Obs.	i.i.d. null		block-21 null		Verdict
		mean	5–95 %	mean	5–95 %	
H1 H^{dep} gap	−0.99	−0.21	−0.42–−0.00	−0.51	−0.90–−0.17	clears both
H1 $\mathcal{J}$ gap	1.93	2.14	1.59–2.79	1.97	1.18–2.90	fails both
H1 $\mathcal{D}$ gap	0.27	1.25	0.82–1.75	0.65	0.20–1.19	against (i.i.d.)
H2 $\text{corr}(\Delta H^{vol}, \Delta H^{dep})$	0.10	0.23	−0.26–0.60	0.27	−0.21–0.63	fails both
H2 β (stress)	0.20	0.05	−0.33–0.31	0.11	−0.28–0.32	fails both
H3 upward jumps	158.0	148.5	136.9–161.0	133.2	121.0–146.0	block only
H3 downward jumps	9.0	3.0	1.0–6.0	3.7	1.0–7.0	against (both)
H3 mean upward jump (nats)	0.62	1.12	0.95–1.29	0.84	0.69–1.03	against (both)
H3 skewness of $\Delta\mathcal{J}$	12.26	14.80	12.27–17.60	12.97	9.71–17.65	against (i.i.d.)
H3 var. share of up-jumps	0.838	0.899	0.869–0.922	0.817	0.764–0.860	against (i.i.d.)
H4 half-life (days)	120.8	92.1	82.4–103.3	162.8	116.0–227.8	between the two nulls
H4 $\text{sd}(\mathcal{J})$	2.03	1.55	1.31–1.81	2.06	1.40–2.75	i.i.d. only
H4 $\text{range}(\mathcal{J})$	18.95	10.05	8.05–12.88	13.77	9.04–17.88	clears both
H5 ΔS_{mkt} gap	−0.053	−0.011	−0.088–0.070	−0.027	−0.082–0.039	fails both
H5 $\text{corr}(\Delta\mathcal{J}, \Delta S_{mkt})$	−0.243	−0.035	−0.298–0.230	−0.069	−0.268–0.209	fails both
H6 coupling, stress	0.63	0.90	0.84–0.94	0.72	0.56–0.82	against (i.i.d.)
H6 coupling gap	0.590	0.769	0.671–0.871	0.668	0.513–0.822	against (i.i.d.)
– $\text{corr}(\mathcal{J}, \log \text{vol})$	0.461	0.414	0.359–0.466	0.264	0.164–0.362	—

Table 2: S&P 500: observed statistics against two calibrated nulls, two hundred replications each. “Clears” means that the share of replications at least as extreme as the observed value, on the side predicted by the hypothesis, is at most 5%; “against” means that this threshold is reached on the opposite side, i.e. the null exceeds the data. Gaps are stress minus calm. Exact p -values are in `paper/tables/sp500_placebo_null*.csv`.

What clears both nulls. Two things, and only two. The regime gap of the *dependence* channel: -0.99 nat against -0.21 (i.i.d., none of the two hundred replications reaches it) and -0.51 (block, $p = 0.030$: six replications out of two hundred are at least as extreme, so the result is more fragile against this second null). And the *range* of the structural index: 18.95 nats against 10.05 (i.i.d., no replication reaches it) and 13.77 (block, $p = 0.035$: the result is more fragile against this second null, whose 95% quantile is 17.88). In other words: what separates a crisis from a mere run of large returns is that the cross-section couples, and the structural index travels further in the real market than in any resampling of the same returns.

What correcting the stress threshold removed. In an earlier version of this project, the gap in $\mathcal{J}$ cleared the i.i.d. null (2.89 against 2.18 , $p = 0.02$). With a stress threshold computed on the past only, the gap falls to $+1.93$ nat, less than the mean of both nulls (2.14 and 1.97 , $p = 0.67$ and 0.50): it no longer clears either. The dependence-channel gap has also shrunk (-1.72 before, -0.99 now), but consistently in the data and in the nulls, and it still clears both.

What does not clear. Jump asymmetry fails the i.i.d. null outright (H3). Channel coupling fails both, and fails in the direction that matters: the i.i.d. null couples *more* strongly than the data ($p = 0.01$), and so does the block null on average, without the difference being clear-cut (H6). The regime signal of the shape channel ($+0.27$ nat) is smaller than that of the i.i.d. null ($+1.25$) and, without the difference being clear-cut, than that of the block null ($+0.65$, $p = 0.08$). The offset hypothesis is lost in the nulls' own noise (H2).

What is not identified. The relaxation half-life is 121 days against 92 under the i.i.d. null and 163 under the block null — the data sit *between* the two, and the sign of the comparison flips (only 8.5% of block replications fall below 121 days, against no i.i.d. replication above). Block resampling sometimes concatenates high-volatility blocks and manufactures long excursions; i.i.d. resampling manufactures short ones. The honest conclusion is that the level of the half-life is an artefact of which null one chooses to believe, and that P1 is supported by the amplitude result rather than by the half-life. The skewness signature (H5) has the right sign against both nulls but does not pass the 5% threshold ($p = 0.10$ and 0.08 for the correlation between $\Delta\mathcal{J}$ and ΔS_{mkt}), which is suggestive and no more.

A remark on $\mathcal{J}$ and volatility. Under the block null, the correlation between $\mathcal{J}$ and log-volatility is 0.26 ; in the data it is 0.46 ($p < 0.005$). The structural index tracks volatility *more* closely than a clustering-only process, not less. Table 7 shows that the shape channel nevertheless stays orthogonal to volatility: the extra tracking therefore goes through the dependence channel — correlation and volatility genuinely move together in a way that clustering alone does not reproduce.

The same test on industry portfolios. Table 3 reproduces exactly the pipeline and both nulls (two hundred replications each) on Kenneth French's 48 daily industry portfolios, 1990–2026, with the same look-ahead-free stress threshold. This is the first time this panel is confronted with the nulls. The regime gap of the *dependence* channel clears both: -3.65 nats against -0.56 (i.i.d., no replication) and -2.46 (block, $p = 0.025$), about six times and one and a half times the null. The magnitude relative to the nulls is similar on single stocks and on industries (4.7 and 1.9 times the null), which argues against an industry-aggregation artefact. The gap in $\mathcal{J}$, on the other hand, no longer clears anything ($+5.93$ against 7.95 and 7.79 ; it cleared both before the threshold correction) and the rise of the deficit $\mathcal{D}$ under stress is below that of both nulls (0.98 against 4.81 and 2.60).

The *range* of $\mathcal{J}$ clears only the i.i.d. null (54.5 against 36.9) and not the block null (45.9, $p = 0.13$): the second result of the main panel does not replicate here. The half-life (190 days) again sits between the two nulls (101 and 199), the count of upward jumps (182) is that of both nulls (182 and 176), and channel coupling under stress stays below that of the i.i.d. null. A remark for H2: the correlation of -0.63 between ΔH^{vol} and ΔH^{dep} , which the earlier version of the paper presented as the industry offset, is approached by the block null (-0.54 , $p = 0.09$); it clears only the i.i.d. null (-0.36).

Statistic	Obs.	i.i.d. null		block-21 null		Verdict
		mean	5–95 %	mean	5–95 %	
H1 H^{dep} gap	−3.65	−0.56	−1.03–−0.19	−2.46	−3.45–−1.41	clears both
H1 $\mathcal{J}$ gap	5.93	7.95	6.12–10.11	7.79	5.37–10.39	against (i.i.d.)
H1 $\mathcal{D}$ gap	0.98	4.81	3.47–6.39	2.60	1.15–3.99	against (both)
H2 $\text{corr}(\Delta H^{vol}, \Delta H^{dep})$	−0.63	−0.36	−0.54–−0.19	−0.54	−0.65–−0.41	i.i.d. only
H2 β (stress)	0.23	0.08	−0.05–0.21	0.28	0.15–0.41	i.i.d. only
H3 upward jumps	182.0	181.8	165.0–201.1	176.5	159.0–194.1	fails both
H3 downward jumps	3.0	7.8	4.0–13.0	5.2	2.0–10.0	i.i.d. only
H3 mean upward jump (nats)	1.70	3.47	2.74–4.18	2.30	1.82–2.90	against (both)
H3 skewness of $\Delta\mathcal{J}$	17.04	18.31	14.84–21.83	16.46	11.76–26.04	fails both
H3 var. share of up-jumps	0.889	0.943	0.918–0.958	0.877	0.840–0.910	against (i.i.d.)
H4 half-life (days)	189.8	101.0	89.7–112.1	199.3	136.6–273.6	between the two nulls
H4 $\text{sd}(\mathcal{J})$	6.37	5.46	4.43–6.48	7.05	5.05–9.18	fails both
H4 $\text{range}(\mathcal{J})$	54.47	36.86	28.53–43.95	45.86	32.98–59.10	i.i.d. only
H5 ΔS_{mkt} gap	−0.112	−0.057	−0.140–0.017	−0.068	−0.123–−0.010	fails both
H5 $\text{corr}(\Delta\mathcal{J}, \Delta S_{mkt})$	−0.440	−0.197	−0.472–0.075	−0.177	−0.441–0.149	fails both
H6 coupling, stress	0.76	0.94	0.91–0.96	0.82	0.74–0.87	against (i.i.d.)
H6 coupling gap	0.411	0.635	0.510–0.757	0.558	0.390–0.717	against (i.i.d.)
– $\text{corr}(\mathcal{J}, \log \text{vol})$	0.677	0.443	0.388–0.490	0.327	0.240–0.409	—

Table 3: FF49, 48 industry portfolios, 1990–2026: observed statistics against the two calibrated nulls, two hundred replications each. Same conventions as Table 2. Exact p -values are in `paper/tables/ff49_placebo_null*.csv`.

And one thing none of this touches. The stress-testing construction of Section 8 is a measurement convention plus a convex optimisation. It does not rest on the truth of any of H1–H7.

7.2 H1: the regime signature

Table 4 compares dates on which rolling 21-day market volatility exceeds its historical decile, computed on the past only (1,076 dates), with the rest (6,481); the first 504 dates, without history, are excluded. Block-bootstrap p -values summarise in one line a null in which the regime label carries no information about the level.

Every sign is the one H1 predicts. Dependence entropy falls by 0.99 nat (1.66 with the EWMA estimator), the effective number of independent risk modes goes from 13.1 to 11.4 out of 20, the shape deficit rises by 0.27 nat (not significant, $p = 0.16$), and the structural index rises by 1.94 nat — a 38% increase, the most significant difference in the table. The auxiliary panel agrees on the signs of dependence and of $\mathcal{J}$ (from 0.92 to 1.15 nat, $p = 0.034$) but its sample, cut by the burn-in

Channel	Stress	Calm	Difference	p
H^{vol} (scale)	-75.56	-81.53	+5.97	< 0.001
H^{dep} (dependence)	-4.50	-3.50	-0.99	0.020
H^{dep} (EWMA version, used in $\mathcal{J}$)	-5.36	-3.70	-1.66	< 0.001
H^{cov} (Gaussian ceiling)	-80.06	-85.03	+4.97	< 0.001
$\mathcal{D}$ (shape)	1.72	1.45	+0.27	0.156
skewness channel	0.120	0.110	+0.009	0.345
tail channel	1.604	1.342	+0.262	0.160
$\mathcal{J}$ (structural index)	7.09	5.15	+1.94	< 0.001
mean pairwise correlation	0.320	0.256	+0.065	0.021
effective number of modes	11.42	13.07	-1.65	0.020
tail co-exceedance rate	0.274	0.221	+0.053	0.020
market skewness	-0.216	-0.233	+0.017	0.808
market excess kurtosis	2.20	1.73	+0.47	0.306

Table 4: H1: channel means by regime, look-ahead-free stress threshold, individual S&P 500 stocks, 1990–2022. Entropies in nats. The dependence channel is reported for both covariance estimators; the nulls of Table 2 concern the first.

period to 1,104 dates of which 160 are in stress, no longer carries a significant dependence difference (-0.19 , $p = 0.17$).

Against the calibrated null, the hypothesis splits in two. The *dependence* channel carries a real signal: the observed regime gap of -0.99 nat is about five times the -0.21 of the i.i.d. null and twice the -0.51 of the block null; it lies outside all two hundred replications of the first and all but six of the second. The gap in $\mathcal{J}$ itself, which cleared the i.i.d. null before the threshold correction, no longer clears either null (1.93 against 2.14 and 1.97): it is the dependence component that carries H1, not the aggregate index. The *shape* channel does not: its observed regime gap of $+0.27$ nat is below the $+1.25$ of the i.i.d. null (and the $+0.65$ of the block null). Shuffled data, in which a “stress” window is by construction a window containing isolated extreme returns, fatten the measured marginals more than real stress episodes do. The rise of $\mathcal{D}$ under stress is therefore evidence of nothing beyond the presence of large returns.

The economically meaningful part of H1 is thus narrower than the raw table suggests, and it is the part one would want: what distinguishes a real crisis from a run of large returns is that *the cross-section couples* — the effective number of independent risk factors collapses — not that the marginals fatten.

The Forbes & Rigobon (2002) bias. The stress regime is defined by market volatility, and a correlation measured on a high-volatility subsample is mechanically higher, even when the underlying correlation does not change. Part of the raw gap of Table 4 is therefore this selection bias. Both nulls contain it by construction: they resample whole rows of the return matrix, hence a joint law of same-day returns that does not vary over time, and they recompute the regime with the same volatility filter. In a resampled world, the dependence-channel gap between regimes is *only* the selection effect, and that is what the -0.21 and -0.51 of Table 2 measure. The observed excess is what remains beyond it. The nulls do not cover everything, and we say so: neither is a process with persistent volatility and constant conditional correlation (CCC-GARCH type), which would be the sharpest null for this question; the block null is only an approximation to it, and it is against the block null that the result is most fragile ($p = 0.030$). Nor have we applied the analytical correction of Forbes & Rigobon (2002), which assumes a one-factor model.

Two entries of the table are as instructive as the confirmations. The *level* of market skewness barely moves between regimes ($p = 0.81$): stress is not a state in which returns are persistently more left-skewed, a point we return to with H5, where the skewness signature appears at the time of the jump rather than throughout the regime. And the Gaussian ceiling H^{cov} *rises* under stress, by 4.97 nats — because the scale channel dominates it. Differential entropy computed from a covariance matrix *increases* in a crisis. Any attempt to use the level of Gaussian entropy as a stress indicator therefore has the sign backwards; that is precisely why the paper works with the scale-free $\mathcal{J}$.

7.3 H2: the offset is real, partial, and confined to stress

An earlier version of this project, on the 49 Fama–French industry portfolios, found a correlation of -0.63 between ΔH^{vol} and ΔH^{dep} over 1990–2026 and a 57% reduction in the variance of ΔH^{cov} : marginal volatility and dependence offset each other to the point of leaving the covariance determinant markedly more stable than either component. Under the binary formulation of the time, this result did not replicate on individual stocks and the matter stopped there. Estimating (12) says something more precise.

Offset rate $\hat{\beta}$ (95 % CI)	All	Calm	Stress
S&P 500, 20 individual stocks	+0.09 (−0.02; +0.19)	−0.00 (−0.14; +0.14)	+0.20 (+0.08; +0.32)
EuStockMarkets, 4 indices	+0.33 (+0.23; +0.42)	+0.30 (+0.16; +0.44)	+0.39 (+0.31; +0.47)
R^2 of (12), S&P 500	0.02	0.00	0.16
R^2 of (12), EuStock	0.24	0.17	0.44
FF49 industry portfolios (recomputed): +0.38 all dates, +0.51 calm, +0.23 stress, all significant.			

Table 5: H2: offset rate β estimated by (12), HAC standard errors with Newey–West lags. $\beta > 0$ means the two channels move in opposite directions. Values in bold are significantly positive.

Three readings that the binary formulation prevented.

The sign. Outside the stress regime, β is not distinguishable from zero on the S&P 500: the confidence interval contains 0 and the point estimate is zero in calm markets. **In the stress regime, however, β is significantly positive on the S&P 500** (+0.20, interval +0.08 to +0.32). On **EuStockMarkets**, after excluding the first two years for lack of history for the stress threshold, β is positive in all three regimes (+0.30 calm, +0.39 stress): the calm/stress contrast there is less sharp than it was read, and this small panel (four indices, 1,104 dates) cannot decide. On the main panel, H2 is supported as a regime-conditional hypothesis and rejected as an unconditional one, which $\beta_{\text{stress}} > \beta_{\text{calm}}$ predicted.

The magnitude. It is modest. A β of 0.2 to 0.4 means that only between a fifth and two fifths of a scale-entropy move is absorbed by dependence; the rest passes into the ceiling. This is far from the $\beta = 1$ that a genuine “conservation of covariance volume” would require. The old formulation, by speaking of ceiling stability, implicitly suggested β close to 1: that was too strong even where the phenomenon exists.

The fit. The R^2 of (12) goes from 0.02 to 0.16 on the S&P 500 and from 0.24 to 0.44 on the European indices when restricted to stress. The postulated linear relation is not only of the right sign under stress, it is also much better defined there.

Industry portfolios. On French’s 48 industries, β is significantly positive in all three regimes, and *larger in calm markets* (+0.51) than under stress (+0.23): the offset there is a permanent property of industry aggregation rather than a crisis phenomenon, consistent with the idea that aggregating firms into industries imposes common-factor dependence. It clears only the i.i.d. null (stress β of +0.23 against 0.08) and not the block null (0.28).

A caveat, and a question settled. HAC standard errors remain optimistic on rolling windows that share 251 of their 252 days, and the Newey–West rule (10 lags here) is more so than the fixed 21 lags of the earlier version; the intervals of Table 5 should be read as indicative. As to whether β_{stress} survives the calibrated null of Section 7.1, the question is now settled, and in the direction this paragraph feared: **no**. Table 2 now includes this test. Under the i.i.d. null, the observed β_{stress} of 0.20 falls inside the null distribution (0.05, interval -0.33 – -0.31 , $p = 0.40$); under the block null, the same verdict (0.11, interval -0.28 – -0.32 , $p = 0.47$). The same mechanical reason that lets resampling reproduce H6 — turbulent windows containing large common days — suffices to produce a positive β slope under stress, without any economic offset existing in the resampled data. **H2 in the form (12) is therefore a measured regularity and not an established result:** the sign and order of magnitude reported above are real, but a resampling with no offset dynamics produces an equally positive β_{stress} , for the same mechanical reason as H6. The verdict thus joins the one already reached on the correlation.

What this does not change. The consequence that motivated Section 3 stands: with β of order 0.2 outside crises and a Gaussian ceiling that rises with volatility, the information content of a crisis cannot be read in $\det \Sigma$. It has to be sought elsewhere.

7.4 H3: the jump component is one-sided

Taken at face value, the evidence looks overwhelming. On the main panel, the four-sigma detector finds 158 upward jumps against 9 downward jumps; upward jumps carry 83.8% of the variance of $\Delta \mathcal{J}$ against 0.3% for downward jumps; the skewness of $\Delta \mathcal{J}$ is +12.3. The auxiliary panel agrees (27 against 5, skewness +6.7). Figure 1 is visibly a sawtooth, with sharp rises in 1997–98, 2002, September 2008, 2011, 2015, 2018 and a 21-nat peak in March 2020, each followed by a slow decay — exactly the shape (11) predicts.

The asymmetry does not survive the i.i.d. null. Under that null, the detector finds *on average 148 upward and 3 downward jumps*, with a skewness of $\Delta \mathcal{J}$ of +14.8 — above the observed +12.3 — and an upward-jump variance share of 90%, again above the observed value. Every one-sidedness statistic of the data lies at the edge of or inside this null.

The block null is more lenient on one point and not on the others. Preserving volatility clustering lowers the null’s count of upward jumps to 133.2, so the observed 158 lies well beyond it ($p = 0.01$); but the skewness of $\Delta \mathcal{J}$ (13.0) and the upward-jump variance share (0.82) still bracket the observed values, and the mean upward jump remains larger in the null than in the data. A count that clears one null while every measure of the *shape* of the asymmetry fails both is not a result. **H3 is not supported.**

The explanation is not subtle once stated. $\mathcal{J}$ is a positive convex functional of estimated second, third and fourth moments. A single fat-tailed observation entering an exponentially weighted estimator pushes it up sharply, and the estimator’s own memory then lets it decay smoothly. Sawtooth dynamics are what a fat-tailed marginal law produces when passed through this measurement apparatus, whether or not information arrives in bursts. The visual appeal of Figure 1 is, to that extent, an artefact.

Two things are worth saving. First, the null is demanding and probably too demanding for the postulate as stated: because i.i.d. resampling preserves fat tails, it preserves the *bursts* of information and scrambles only their chronology. What the test rejects is therefore not “information arrives in discrete lumps” but the weaker and more useful claim that *the observed one-sidedness of $\Delta\mathcal{J}$ carries information about arrival dynamics beyond what the marginal law already implies*. It does not.

Second, one comparison in Table 2 goes the other way and is instructive. The largest observed upward jump is 3.21 nats against a mean of 6.86 under the i.i.d. null, and the mean upward jump 0.62 against 1.12 (i.i.d.) and 0.84 (block): real entropy jumps are about *half* the size the shuffled world produces. The reason is that real extreme days arrive when the market is already turbulent, so the local scale against which they are measured is already high; in the shuffled world they spring from a calm baseline and look all the more surprising. This is volatility clustering showing up in the entropy measure — evidence of temporal structure, obtained from a statistic built to test something else entirely.

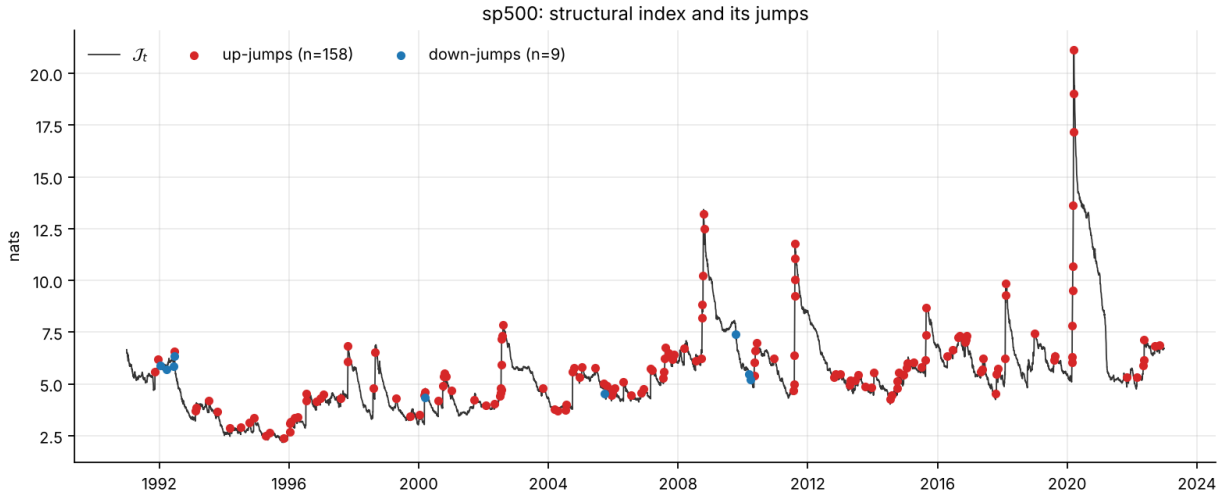


Figure 1: The structural index $\mathcal{J}_t$ on twenty S&P 500 stocks, 1990–2022, with detected jumps. Red points are upward, entropy-destroying jumps; blue points are downward jumps. The sawtooth — sharp rise, slow decay — is the qualitative content of (11).

7.5 H4: relaxation, against the estimator’s own memory

Estimating (11) on non-jump dates only — so that the diffusive channel is identified without contamination — gives $\kappa = 0.0057$ per day ($t = -4.70$, HAC with 10 lags, $p = 0.0001$ under the Dickey–Fuller distribution), a half-life of **121 trading days** and a long-run level of 3.30 nats. The auxiliary panel gives $\kappa = 0.0056$, a half-life of 124 days, $p = 0.047$ under Dickey–Fuller (0.004 under the normal law, which was the value reported until now and which over-rejects).

The half-life is where the two nulls diverge most sharply, and the divergence is instructive. Under i.i.d. resampling, the same estimator returns 92.1 days on average, so the observed 120.8 exceeds every one of the two hundred replications. Under block resampling, it returns 162.8 days on average (90% interval: 116–228), so the observed value falls *below* 91% of replications. The data sit between the two nulls, and the sign of the comparison depends entirely on which one is asked. Block resampling sometimes concatenates high-volatility blocks and manufactures long excursions; i.i.d. resampling breaks the runs that real markets contain and manufactures short ones. The level of the

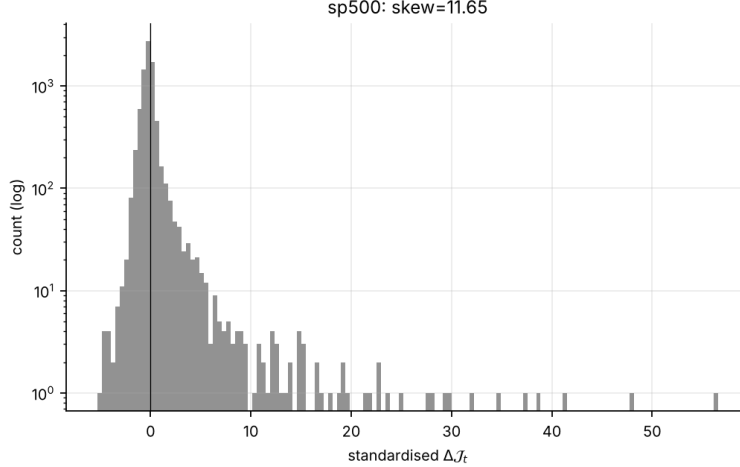


Figure 2: Distribution of standardised $\Delta\mathcal{J}_t$ (counts on a log scale). The right tail extends well beyond the left.

half-life is therefore **not identified**, and the figure of 121 days must not be read as an absorption time — the controlled Gaussian simulation of Section 6.3, in which a single shock decays in 35 days without any economics, sets the floor of what belongs to estimator memory.

What survives is the *amplitude*. The range of $\mathcal{J}$ over the sample is 18.95 nats against 10.05 under the i.i.d. null (none of the two hundred replications reaches it) and 13.77 under the block null ($p = 0.035$; 95% quantile of the null: 17.88). Its standard deviation clears the i.i.d. null but not the block null. Real markets move the structural index further than any resampling of the same returns does, which is the part of P1 the data support: not a particular relaxation rate, but a state variable that genuinely travels.

Figure 3 adds a caveat that the point estimate hides. Averaging the index around detected upward jumps shows that $\mathcal{J}$ starts rising some twenty days *before* the flagged date, peaks about thirteen days after, and returns to its pre-event level after roughly eighty days. Information events cluster: a flagged jump is typically not an isolated arrival but one element of a sequence. The clean picture of a single jump followed by decay given by (11) is therefore an idealisation, and a self-exciting Hawkes-type arrival process would fit the observed clustering better than the Poisson arrivals assumed here.

7.6 H5: jumps carry a sign

Postulate P2 predicts that upward jumps come with negative market skewness. On upward-jump dates, the change in market skewness averages -0.052 against $+0.001$ elsewhere; the level of market skewness on those dates is -0.394 against -0.214 ; and $\text{corr}(\Delta\mathcal{J}, \Delta S_{mkt}) = -0.24$. The skewness channel of the deficit rises on jump dates ($+0.010$ against -0.0002), as does tail co-exceedance ($+0.008$ against -0.0002 , correlation with $\Delta\mathcal{J}$ of $+0.35$). The auxiliary European panel is stronger on this dimension: $\text{corr}(\Delta\mathcal{J}, \Delta S_{mkt}) = -0.38$ and a mean skewness change of -0.129 on upward-jump dates.

The caveat already visible in Table 4 is that the skewness signature is an *event* property, not a regime property: sorting dates by volatility reveals no significant skewness difference, sorting them by entropy jump does.

Against the nulls, H5 is suggestive and nothing more. Both produce the effect in the same



Figure 3: Average path of $\mathcal{J}_t$ around detected upward jumps, re-centred at the event date, with the interquartile band. The rise before day zero is the clustering of jumps.

direction without any economics, because the mechanical link is obvious once stated: a large negative return simultaneously raises $\mathcal{J}$ and lowers the measured skewness of the window it enters. The observed correlation of -0.243 lies between the eighth and tenth percentile of both nulls (-0.035 i.i.d., -0.069 block; $p = 0.10$ and 0.08), and the mean skewness change on jump dates (-0.053) stays inside both (-0.011 and -0.027 ; $p = 0.18$ and 0.22). We report H5 as *not established*: the direction is right, it is consistent across two nulls and two datasets (the auxiliary panel gives -0.38), but a p -value of 0.08 to 0.10 over two hundred replications, hence above the 5% threshold, is a reason to look further, not a result. Postulate P2 remains a postulate.

7.7 H6: the channels lock together under stress

$\text{corr}(-\Delta H^{\text{dep}}, \cdot)$	All	Calm	Stress
S&P 500: with $\Delta \mathcal{D}$ (shape)	0.35	0.04	0.63
S&P 500: with the tail channel	0.35	0.02	0.64
S&P 500: with the skewness channel	0.06	0.17	0.01
EuStock: with $\Delta \mathcal{D}$	0.06	-0.05	0.36

Table 6: H6: coupling between the two entropy-destroying channels, by regime.

In the data, the pattern is striking. In calm markets, the dependence channel and the tail channel are effectively unrelated (0.02); under stress their correlation is 0.64 . The auxiliary panel reproduces it at -0.05 against 0.36 , and the coupling is carried entirely by the *tail* channel, the skewness channel taking no part.

Both nulls reproduce it too, and the i.i.d. null reproduces it *more strongly* than the data: a stress coupling of 0.90 (i.i.d.) and 0.72 (block) against the observed 0.63 , and a stress-minus-calm gap of 0.77 and 0.67 against the observed 0.59 . Only the i.i.d. null clearly exceeds the data ($p = 0.01$ for the gap); against the block null, the observed value is inside the distribution ($p = 0.20$). The mechanism is mechanical rather than economic. A window containing a large common return shows both a higher measured correlation and fatter measured marginals, so the two channels move together in

any data where large days exist, shuffled or not. H6 is **not supported** as evidence of a binding common constraint.

Adding the block null does change the interpretation of the gap between null and data. Under i.i.d. resampling, the null’s coupling is almost deterministic (0.90, standard deviation 0.03); preserving clustering lowers it to 0.72 with a standard deviation of 0.08, closer to and now compatible with the observed 0.63. Real stress episodes couple the two channels *less* tightly than the i.i.d. null and no more than the block null. If the gap with the i.i.d. null is not noise, it points away from Section 4: dependence and tail weight would have partly independent drivers in real crises, which a common-constraint story does not predict.

Over the full sample, the variance of $\Delta\mathcal{J}$ splits 40% dependence and 60% shape on individual stocks, and 69%/31% the other way round on the four-index panel — a reminder that the balance between channels depends on the cross-section being measured.

7.8 H7: entropy describes, it does not forecast

The question, in one sentence. If I know the market’s entropy today, do I know something about the risk of the next three weeks that today’s volatility did not already tell me?

The protocol, in four steps. Two models forecasting the same thing are compared — realised volatility over the next 21 days, then drawdown, then the worst daily loss. The *baseline model* uses only the scale channel. The *augmented model* adds an entropy channel. Then:

1. both models are estimated on the past only, never on the future being forecast;
2. a forecast is made and its error recorded;
3. each model is re-estimated every year with the elapsed year added, and the process repeats;
4. errors are compared only on dates 21 days apart, so that two successive forecasts never concern the same days.

What “confirmed” would mean here. Only this: that the augmented model makes, *on dates it had not seen when estimated*, smaller errors than the baseline, and that the gap is too large to be chance — which the test of Diebold & Mariano (1995) decides. The point deserves emphasis, because it is where most indicators of this kind fool themselves: **adding any variable mechanically improves the fit on the data used to estimate the model**. A good in-sample fit proves nothing at all.

The verdict. H7 is rejected. No channel — not $\mathcal{J}$, nor the deficit $\mathcal{D}$, nor the dependence term, nor the crisis indicator, nor the unabsorbed share — significantly improves the out-of-sample forecast. The best improvement observed, across all channels and targets, is 3.3% of mean squared error with $p = 0.23$: indistinguishable from chance. Several specifications are significantly *worse* than the baseline.

And the exact illustration of the trap described above: *in sample*, the coefficients are highly significant, $p < 10^{-4}$ for $\mathcal{J}$ in every specification. This is the pattern one expects from overlapping windows, and it is why these figures are not presented as evidence.

What to conclude. That entropy channels are *diagnostic*, not *predictive*: they describe the market’s present informational state well, they do not anticipate its next move under a linear specification. This result replicates, on another dataset and under a markedly more demanding protocol, the negative predictive result obtained by the earlier version of this project. The value of the framework is therefore to be sought where Section 8 seeks it — in internal consistency and scenario calibration — and not in point forecasts.

7.9 Is all this just volatility?

Correlation with rolling log-volatility	S&P 500	EuStock
H^{vol} (scale channel)	+0.56	+0.45
$-H^{dep}$ (dependence channel)	+0.56	+0.66
$\mathcal{D}$ (shape channel)	+0.08	−0.06
$\mathcal{J}$ (structural index)	+0.46	+0.60
tail co-exceedance	+0.32	+0.49

Table 7: What each channel shares with volatility.

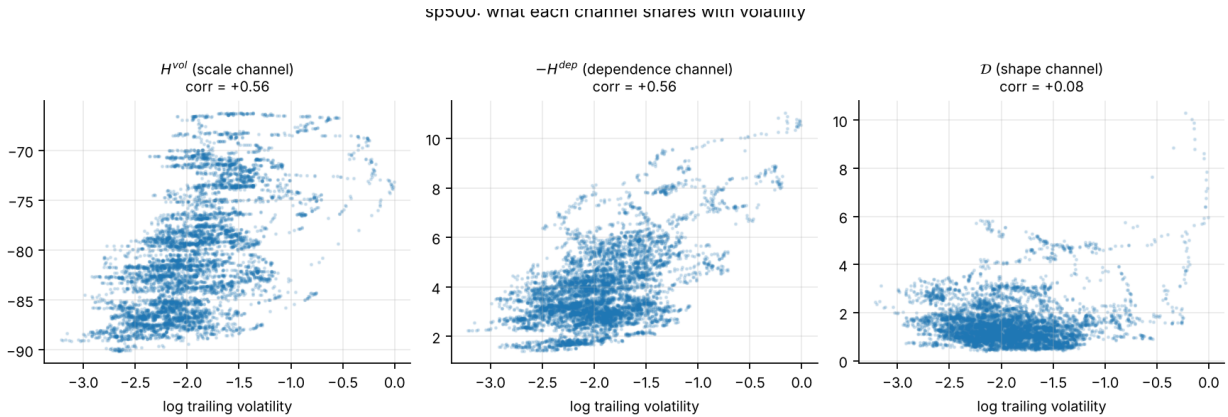


Figure 4: Each channel against rolling log-volatility. The scale and dependence channels track it; the shape channel does not.

Table 7 and Figure 4 answer the obvious objection. The structural index shares about a fifth of its variance with volatility on individual stocks ($r = 0.46$): it is related to it but far from redundant. The shape channel is *orthogonal* to volatility ($r = 0.08$ and -0.06): the tail thickness of the conditional distribution is a dimension of the market state that a volatility measure does not see at all. Since the shape channel carries 60% of the variance of $\Delta\mathcal{J}$ on the main panel, most of what moves the index is information that volatility does not contain.

7.10 The auxiliary European panel

Figure 5 applies the identical pipeline to the four European indices, 1991–1998. The panel is far too small to carry a result on its own — four assets give six pairwise correlations against 190 for the main panel, and a single stress cycle, and it has not been run against the nulls — but the raw signs reported above are found again: $\mathcal{J}$ rises under stress (1.15 against 0.92, $p = 0.034$), the channels couple under stress and not in calm markets (0.36 against -0.05), the index relaxes with a half-life

of 124 days, and upward jumps carry a mean change in market skewness of -0.129 against $+0.003$ elsewhere.

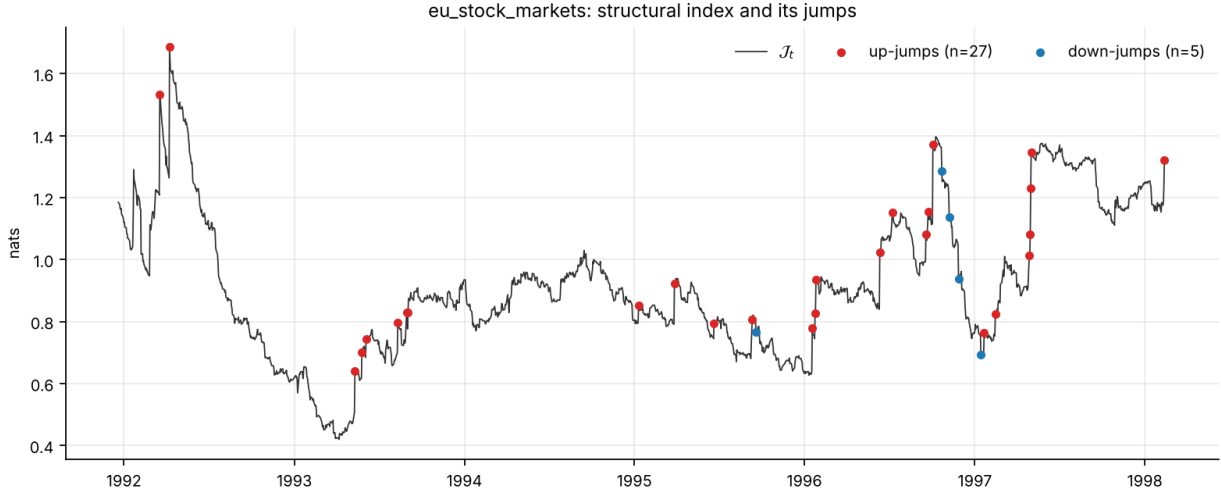


Figure 5: Auxiliary panel: the structural index on the DAX, SMI, CAC and FTSE, 1991–1998. The visible rises coincide with the ERM episodes of September 1992 and 1993 and with the Asian crisis of October–November 1997.

8 Entropy stress testing

8.1 The construction

Definition 2 (Entropy-ball scenario). Given today’s conditional distribution p_0 and a severity $\eta > 0$, the scenario of severity η for a risk functional ρ is

$$p_\eta^* = \arg \max_{p: D_{KL}(p \| p_0) \leq \eta} \rho(p). \quad (13)$$

For a linear portfolio and a Gaussian reference, only the one-dimensional projection matters. With m_0, s_0 the portfolio mean and standard deviation and $u = (m - m_0)/s_0$, $v = s/s_0$, the constraint is $g(u, v) = \frac{1}{2}(u^2 + v^2 - 1 - 2 \log v) \leq \eta$, and for any ρ of the form $-m + cs$ — Gaussian VaR and expected shortfall are both of that form — the solution satisfies $u = -t$, $v - v^{-1} = ct$, where t is pinned down by $g = \eta$. The special case $c = 0$ is the sanity check to remember:

$$\text{a purely directional scenario of } \eta \text{ nats is exactly a move of } \sqrt{2\eta} \text{ sigmas.} \quad (14)$$

One nat buys 1.41 sigma. This is the exchange rate between information and risk, and it is the whole reason the construction is usable by a committee: the severity parameter is not an abstraction.

8.2 Calibrating the radius

The radius is not a modelling choice here. Define the h -step information flow

$$I_t^{(h)} = D_{KL}(p_t \| p_{t-h}), \quad (15)$$

the divergence between the market’s conditional distribution today and h days ago — how much the market has revised its own view. The divergence must, however, be measured on the same object as the constraint. The ball (13) of the previous section constrains only the *portfolio projection*: the law of $w^\top r$, in one dimension. The radius must therefore be read off the information flow of that same projection, $I_t^{(h)} = D_{KL}(p_t^w \parallel p_{t-h}^w)$, and not off that of the whole return vector, which grows with the number of assets and overstates the projection. An earlier version of this project calibrated the one-dimensional ball on the N -dimensional flow: its radii were about ten times too large at the 10-year rung (3.88 nats instead of 0.41), which produced a 10-year volatility multiplier of 3.2 instead of 1.6. Quantiles of $I^{(21)}$, thinned into non-overlapping blocks, convert a return period into a radius. The thinning depends on its starting phase (one date in 21 is kept): the reported radius is the median of the quantiles over the 21 possible phases, and their range is given in Table 8.

Return period	1 yr	2 yr	5 yr	10 yr	20 yr	50 yr [†]
radius η (nats)	0.017	0.045	0.149	0.408	0.634	1.369
range across phases [§]	0.015–0.019	0.034–0.063	0.111–0.187	0.291–0.604	0.472–0.745	0.683–2.082
directional shock (σ)	−0.08	−0.13	−0.24	−0.38	−0.47	−0.66
volatility multiplier	1.12	1.20	1.37	1.63	1.80	2.21
annualised portfolio vol.	22%	24%	27%	33%	36%	44%
one-day ES ₉₉	3.9%	4.2%	4.9%	5.9%	6.6%	8.3%
<i>realised</i> : volatility ratio [‡]	1.54	1.77	2.28	3.25	3.57	4.43

Table 8: Severity ladder for an equally weighted portfolio of the twenty stocks, calibrated on the 21-day information flow of the portfolio projection (383 non-overlapping observations). Today’s reference is an annualised volatility of 20% and a one-day ES₉₉ of 3.4%. [†] beyond the sample: the 50-year rung requires a quantile that 33 years of history cannot resolve and is an extrapolation of the empirical tail (**extrapolated** in `stress_scenarios.csv`). [§] Minimum and maximum over the 21 thinning phases. [‡] Ratio of the portfolio’s realised volatility over the next 21 days to its volatility over the previous 252 days, quantile of the same return period (`scripts/ladder_plausibility.py`).

Three features of Table 8 deserve careful reading. The scenario spends its budget mostly on *variance* rather than direction — at the 10-year rung, 63% more volatility and a drift shock of 0.38σ — because expected shortfall is more sensitive to scale than to location. The radius is not comparable from one portfolio to another: the four-index panel runs from 0.02 to 0.07 nat where the twenty-stock panel runs from 0.02 to 1.4. A ladder must be calibrated for the portfolio to which it will be applied.

The third feature is a limitation, and a serious one. The last line of the table confronts the ladder with history. A 10-year event in the table multiplies portfolio volatility by 1.6; yet the 10-year quantile of the realised 21-day volatility increase is 3.2 (3.0 on French’s industries). At the one- to twenty-year rungs, the scenario’s multiplier is only between half and three quarters of the realised ratio. Two readings are possible, and we do not decide between them: the information flow is measured on conditional laws with exponentially weighted covariance, which react slowly and understate the effective revision of the law over a month; or the Gaussian ball of radius η is simply not the right geometry for volatility shocks this abrupt. Either way, the return periods attached to the scenarios must not be quoted as such: the ladder is a consistent, reproducible calibration rule, not a validated measure of crisis frequency.

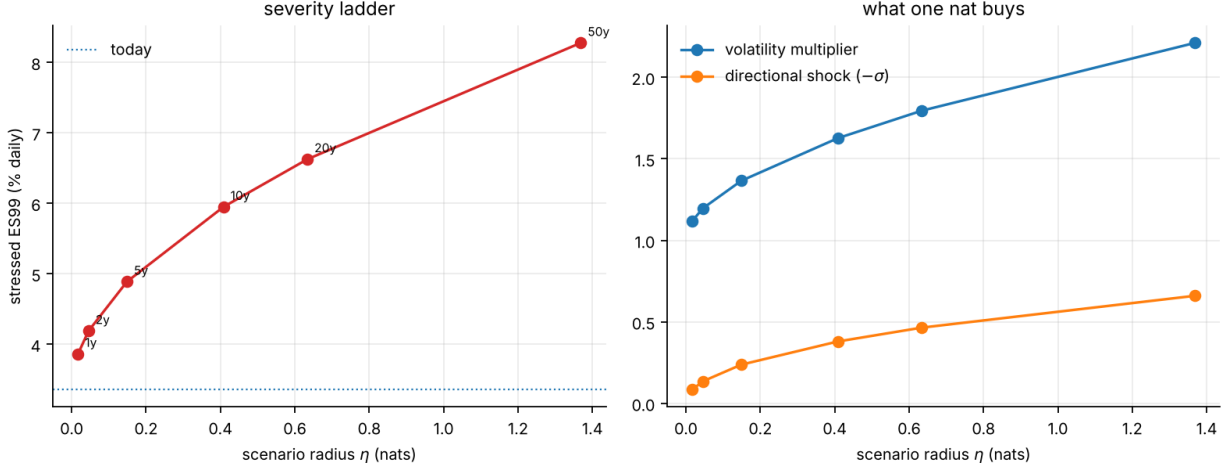


Figure 6: Left: one-day ES99 under stress as a function of the scenario radius, with return periods marked. Right: what one nat buys, split between the volatility multiplier and the directional shock.

8.3 What this adds to volatility, mean and skewness stress tests

The comparison is not rhetorical, because the same metric prices a classical scenario. Take today’s conditional distribution and build the textbook bundle: marginal volatilities multiplied, all pairwise correlations forced to a common high number, a directional shock of several sigmas. Its information price is $D_{KL}(p_{\text{classical}} \| p_0)$, computable in closed form.

Table 9 is the applied point of the paper, and what it establishes is narrower than was once written. What holds: for the same tail loss, the textbook bundle costs more than it needs to. The standard bundle delivers an expected shortfall of 13.7%; an entropy scenario delivers it for 5.47 nats, where the portfolio pays 7.24 nats for the bundle, an overpayment of 24% (33% for the moderate bundle, 15% for the severe; 15, 11 and 6% on French’s industries). The reason is structural: the bundle also deforms directions the portfolio does not see. What no longer holds: the reading “the committee thinks it specified a thirty-year event and bought a ten-year one”, which compared a twenty-dimensional price to a projection radius. On the consistent portfolio scale, all three bundles lie beyond the largest radius observed in 33 years, and the limitation of the previous paragraph forbids turning that into a return period: one can say they are overpaid, not that they are thirty-year events.

Where does the money go? The directional component is cheap and exactly priced: moving the portfolio mean by k standard deviations costs $k^2/2$ nats, so 4.5 for three sigmas. The expensive component is the uniform doubling of every marginal volatility, at 16.1 nats on its own — more than the whole bundle. Forcing all correlations to 0.9 costs 8.1 nats on its own.

The components are not additive, and the way they fail to be is instructive: the three sum to 28.7 nats while the bundle costs 9.97. Forcing correlations up collapses the covariance onto a near-rank-one subspace to which the reference distribution already grants generous variance — the market mode — so scaling that collapsed structure is much cheaper than scaling the full-rank structure ($\text{tr}(\Sigma_0^{-1}\Sigma_1)$ drops from 20 to 10.5 when correlations are forced, before the volatility multiplier is even applied). The volatility shock and the correlation shock partly pay for each other.

This is the structure described by the dependence channel of H1, the only regime result to clear both nulls. Real crises do not equalise correlations and do not inflate every volatility independently; they load the cross-section onto one or two collective modes. A framework that shocks volatility,

Classical bundle	moderate	standard	severe
volatility multiplier	1.5	2.0	3.0
all correlations forced to	0.70	0.90	0.95
directional shock	-2σ	-3σ	-4σ
implied one-day ES_{99}	9.1%	13.7%	20.3%
ambient price (nats)	6.80	9.97	20.19
of which: directional component	2.00	4.50	8.00
volatility component	4.39	16.14	58.03
correlation component	2.43	8.10	13.50
return period of the ambient price	~ 11 yr	> 30 yr	out of sample
price seen from the portfolio (nats)	2.75	7.24	16.22
entropy scenario at the same ES_{99}	1.85	5.47	13.74
information overpayment of the bundle	33%	24%	15%
position on the portfolio ladder	out of sample	out of sample	out of sample

Table 9: Pricing the textbook stress bundles, and comparing each with the entropy scenario delivering the same expected shortfall. The *ambient price* is the divergence $D_{KL}(p_{\text{classical}} \| p_0)$ on the twenty stocks, read on the twenty-dimensional flow; the *price seen from the portfolio* is the same divergence on the law of $w^\top r$, the only quantity comparable to the radius of a projection ball and to the ladder of Table 8, whose in-sample maximum is about 2 nats (50-year rung: 1.37). The worst single day in 33 years prices at 9.03 nats (reverse stress test, one-day horizon).

correlation and shape as three separate scalars has no way of representing this, nor any way of noticing that its bundle buys severity with a 15 to 33% margin.

Four concrete advantages over the volatility/mean/skewness approach follow.

1. **A single severity scale, in a unit that does not depend on the asset.** Scenarios become comparable across desks, portfolios and asset classes. Severity could be quoted with a return period rather than an adjective, once the ladder is validated against history, which it is not here (Section 8.2).
2. **Internal consistency by construction.** Location and scale shocks come from one budget and one optimisation. A bundle that shocks volatility, correlation and skewness independently can describe, and often does describe, a distribution the market could not produce.
3. **Existing scenarios can be audited rather than replaced.** Any regulatory template or committee judgement receives a price in nats, so that the question of whether a stress test is calibrated becomes arithmetic. This is the practical benefit of Table 9.
4. **The reverse stress test comes for free.** Instead of asking what a scenario does, one asks what the cheapest scenario producing a given loss is. On the main panel, the empirical 99% daily loss (3.2%) costs 1.5 nats; the 99.9% loss (7.7%) costs 4.8; the worst day in 33 years (11.5%) costs 9.0. These prices are for a one-day horizon and the empirical law of the loss; they do not read directly on the 21-day ladder of Table 8.

A caveat on the fourth point. Reverse stress tests computed nonparametrically, by tilting the empirical sample, degenerate as the radius grows: at 9.0 nats, the tilted measure has an effective sample size of 1.0, that is, it is the worst single day dressed up as a distribution. The implementation reports the effective sample size for exactly this reason, and the Gaussian-ball construction should be used as soon as it drops below a few dozen.

8.4 Two monitoring indicators

The jump/diffusion split yields two quantities that can be produced daily at negligible cost. The *crisis* indicator is the standardised upward-jump component of $\mathcal{J}$: how much information arrived today, in units of the local diffusive scale, zero when no jump is detected. The *unabsorbed share* indicator is $\mathcal{J}_t$ relative to its rolling maximum: how much of what has already arrived has not yet relaxed. Under P1, it should decay with half-life $\ln 2/\kappa$; a market where it stays close to one appreciably longer is a market where the shock has not been absorbed. Neither improves a volatility forecast (H7); both describe a state that volatility does not.

8.5 Relevance for insurance solvency

Three points connect this to the insurance strand of the wider project. First, an entropy-ball scenario is a natural object for the ORSA: it produces a complete stressed distribution, not a vector of shocks, so it feeds a capital calculation directly and carries a plausibility statement that remains to be validated (see the limitations). Second, the component decomposition of Table 9 concerns the Solvency II standard formula, whose diversification credit is built on a fixed correlation matrix: shocking volatilities and correlations as separate scalars, the usual conservative adjustment, buys a given tail loss at a 15 to 33% information overpayment, because it does not represent the way observed crises redistribute dependence — onto a few collective modes rather than uniformly. Finally, the coherence properties of entropic risk measures (Ahmadi-Javid, 2012; Artzner et al., 1999) make the $\mathcal{J}$ framework compatible in principle with internal-model capital requirements, although nothing here has been tested on liability portfolios and the short, low-frequency panels typical of insurance data would strain every estimator used above.

9 Limitations

The bridge is an argument, not a proof. Section 2.5 lists the failures of linear impact and of sign exchangeability. Nothing empirical below depends on the volume clock, but the framing does.

The deficit is estimated marginally. The shape channel captures marginal non-Gaussianity and Gaussian-copula dependence; nonlinear tail dependence is measured separately rather than integrated, so $\mathcal{J}$ understates the true divergence, by an amount itself largest under stress.

The mechanical null swallows a whole hypothesis. Jump asymmetry is entirely reproduced by i.i.d. data with the same marginal law, and about two thirds of the measured relaxation belongs to estimator memory. The amplitude result survives; the half-life is not identified; the asymmetry result does not survive, and no reader should quote the 158-to-9 figure without the null next to it. The two hundred replications of Table 2 tightened the p -values, and they moved two verdicts: the range of $\mathcal{J}$ now clears the block null only narrowly ($p = 0.035$) and the skewness signature (H5) no longer passes the 5% threshold.

Poisson arrivals are the wrong model. Figure 3 shows clustering that (11) does not accommodate. A self-exciting Hawkes arrival process is the obvious next specification.

Two panels, of modest size. The dependence channel is sensitive to N , the balance between channels shifts with it, and the KL radius is not comparable from one universe to another. Moving from twenty stocks to forty-eight industry portfolios keeps the dependence-channel result but not

the one on the range of $\mathcal{J}$. The two panels do not cover the same period either (1990–2022 and 1990–2026). A broad cross-section of individual stocks, CRSP style, remains to be done before any figure presented here is treated as a constant of nature.

No head-to-head comparison with the absorption ratio. The dependence channel belongs to the family of concentration measures of the correlation matrix (Section 1.1). We have not confronted it, on the same data and against the same nulls, with the absorption ratio of Kritzman et al. (2011); nothing here says it does better.

The severity ladder extrapolates in the tail, and looks too mild. With 383 non-overlapping observations of information flow, rungs beyond about twenty years are interpolations of the empirical tail, not measurements. Fitting a generalised Pareto tail to $I^{(h)}$ would be the disciplined correction. More serious: once the radius is calibrated on the portfolio projection, as the ball requires, the ladder gives a 10-year volatility multiplier of 1.6 against a 10-year quantile of realised volatility increase of 3.2 (Table 8). The return periods attached to scenarios are therefore not validated, and we do not quote them as frequencies; only the relative efficiency (the bundles’ overpayment) and the calibrated composition withstand this defect.

A look-ahead-free stress threshold, but not a choice-free one. The expanding threshold excludes the first two years of each panel, classifies more dates as stress than the full-sample quantile, and strips the gap in $\mathcal{J}$ of its status as a result. It also brings the European panel (four indices, eight years) down to 1,104 dates of which 160 are in stress, where no regime difference of the dependence channel is significant any more. Newey–West HAC lags are shorter than the fixed 21 lags used before, hence more optimistic on rolling windows.

Nothing here forecasts. H7 is rejected on two datasets under a demanding protocol. Any use of these measures should be diagnostic.

The exploratory part clears only one null. H8 is not identified against the block null, at any block length tested between 5 and 126 days (Table 11). The calibrated composition that uses it (Appendix F) starts from a default uniform dilation that is not a minimum of the ambient divergence, and its result is specific to this panel’s correlation structure. A reader who wanted to use H8 to distinguish causally distinct “types” of crisis should treat it as not established.

10 Conclusion

The theoretical contribution is an explicit chain stating, at each link, what is theorem and what is hypothesis. Maximum entropy under first- and second-moment constraints gives a Gaussian; applied in volume time, with traded volume supplying the variance budget, it gives Gaussian returns conditional on volume, and temporal consistency turns that into a subordinated Brownian motion and then a geometric Brownian price. Read backwards, non-Gaussian prices measure the distance from maximal ignorance, and that distance splits into a dependence channel and a shape channel whose sum, the structural index $\mathcal{J}$, is scale-free and positive. Conditioning cannot increase expected entropy, so entropy falls by jumps whose *expected* size is the mutual information of the news; that this entropy recovers by diffusion, and that the fall is one-sided in the direction of losses, are the two postulates the paper adds and tests.

The empirical picture is above all a warning, and we think that is the useful result. Every descriptive claim of the framework is visible in the data: stress raises the structural index (by 1.9 nat), the index is a sawtooth, jumps are one-sided, they carry negative skewness, and the dependence and tail channels couple under stress. Passed through a null that keeps the marginal law and scrambles the chronology, most of it disappears — and several statistics come out *larger* in the shuffled data than in the real data. A positive convex functional of estimated second, third and fourth moments jumps up and decays smoothly as soon as a fat-tailed observation enters the estimator, whether or not information arrives in bursts. This warning applies to any entropy-based crisis indicator, not only to this one; to our knowledge, the applied literature does not state it in this form.

Exactly two statistics clear both nulls on the main panel, and they are the two to keep; on the FF49 industry portfolios only the regime gap of the dependence channel clears both, so that it, and it alone, holds on both panels. What separates a crisis from a run of large returns is that the cross-section couples: the regime signal of the dependence channel is between one and a half and six times that of the nulls, while that of the shape channel is below both. And the structural index travels further over the sample than in any resampling of the same returns. The relaxation rate itself is not identified — it sits between the two nulls — so P1 is supported by amplitude and not by a half-life. Two further results go against the paper’s interest: the volatility–dependence offset obtained on industry portfolios does not survive the move to individual stocks, and no entropy channel forecasts anything out of sample.

The applied contribution is what the framework offers once one stops asking it to forecast. Measuring the severity of a scenario in nats gives stress testing a unit; calibrating that unit on the market’s realised information flow would give it a frequency, were the resulting ladder not milder than history (Section 8.2); and solving for the worst distribution in the resulting ball gives a scenario that cannot be internally inconsistent. Applied in reverse, it prices the scenarios institutions already run: on this sample, a standard textbook bundle buys its tail loss at a 24% overpayment (15 to 33% depending on severity), because it spends most of its budget inflating every marginal volatility uniformly — which is not the path observed crises take.

Appendix F pushes this observation further, on an exploratory basis: it composes a scenario whose dependence share is anchored on H1 and refined by an eighth hypothesis that clears only the i.i.d. null. What remains established fits in two sentences. Entropy-based crisis indicators must be judged against a resampling of the same returns, or most of their signatures are measurement artefacts. And what withstands that test, on individual stocks and on industries alike, is the coupling of the cross-section under stress.

A Units, invariance, and what is computable

The nat. Entropy $H = -\sum_i p_i \log p_i$ is defined only once the base of the logarithm is chosen, and that choice *is* the choice of unit. With $\log_2$ one gets bits (shannons), with $\log_{10}$ dits, and with the natural logarithm *nats*. One nat is the information delivered by an observation that divides the space of possibilities by $e \approx 2.718$, where a bit divides it by two. The conversion is $1 \text{ nat} = 1/\log 2 \approx 1.4427 \text{ bit}$. This whole paper uses the natural logarithm: entropies, deficits, divergences, scenario radii and the structural index are in nats.

Three operational readings, from the most useful to the most abstract. First the exchange rate established in Section 8: a purely directional scenario of η nats is a move of $\sqrt{2\eta}$ sigmas, so k sigmas cost $k^2/2$ nats. Then coding: $D_{KL}(p\|q)$ nats is the expected excess code length when data with law p are encoded with a code optimal for q . Finally hypothesis testing: a divergence of one nat

per observation means that the likelihood ratio in favour of p grows on average by a factor e with each observation.

h depends on units, divergences do not. The differential entropy of a continuous variable satisfies, for any invertible affine transformation,

$$h(aX + b) = h(X) + \log |a|, \quad \text{and in dimension } N, \quad h(AX + b) = h(X) + \log |\det A|. \quad (16)$$

Expressing returns in percent rather than in decimals therefore adds $N \log 100$ to h : the *level* of h , and hence that of H^{vol} and H^{cov} , has no interpretable absolute value. By contrast $D_{KL}(p \parallel q)$, $I(X; Y)$, $\mathcal{D}$ and $\mathcal{J}$ are invariant under any invertible change of variable, the Jacobians cancelling between numerator and denominator of the logarithm. *A figure in nats is comparable across datasets only if it is a divergence.* That is the structural reason why the paper builds a scale-free index.

How is $h(X)$ computed? In the Gaussian case, in closed form (Appendix B). In general, it cannot be without knowing p . Nonparametric estimators exist — the spacings of Vasicek (1976), Kozachenko–Leonenko nearest neighbours, kernel methods — but all degrade quickly beyond three or four dimensions. Estimating a differential entropy in dimension 20 on a 252-day window is out of reach.

The paper therefore *never* computes $h(p)$. It computes $h(q)$ in closed form from $\widehat{\Sigma}$, and the deficit $\mathcal{D}$ through an approximation of the *difference* of marginal entropies. A difference of entropies is dimensionless and estimable; a bare differential entropy is not. The whole architecture — decompose rather than estimate directly — follows from this constraint.

B Gaussian entropy: proof of equation (5)

Step A: entropy of a multivariate Gaussian. For $q = \mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^N$,

$$\log q(x) = -\frac{1}{2} [N \log 2\pi + \log \det \Sigma + (x - \mu)' \Sigma^{-1} (x - \mu)],$$

so

$$h(q) = -\mathbb{E}_q[\log q] = \frac{1}{2} N \log 2\pi + \frac{1}{2} \log \det \Sigma + \frac{1}{2} \mathbb{E}[(X - \mu)' \Sigma^{-1} (X - \mu)].$$

The last expectation follows from the trace trick: a scalar equals its trace, and the trace is cyclic, hence

$$\mathbb{E}[(X - \mu)' \Sigma^{-1} (X - \mu)] = \mathbb{E}[\text{tr}(\Sigma^{-1} (X - \mu)(X - \mu)')] = \text{tr}(\Sigma^{-1} \Sigma) = \text{tr}(I_N) = N.$$

Collecting terms, $h(q) = \frac{1}{2} [N \log 2\pi + \log \det \Sigma + N] = \frac{1}{2} \log((2\pi e)^N \det \Sigma)$. $\square$

Step B: factorising the determinant. With $\Sigma = DRD$, $D = \text{diag}(\sigma_1, \dots, \sigma_N)$ and R the correlation matrix, multiplicativity of the determinant gives $\det \Sigma = \det(D) \det(R) \det(D) = (\prod_i \sigma_i)^2 \det R$. $\square$

Step C: substitution. $\frac{1}{2} \log \det \Sigma = \frac{1}{2} \log[(\prod_i \sigma_i)^2 \det R] = \sum_i \log \sigma_i + \frac{1}{2} \log \det R$, hence $h(q) = \frac{N}{2} \log(2\pi e) + \sum_i \log \sigma_i + \frac{1}{2} \log \det R$, which is equation (5). $\square$

Sign of the dependence channel. Hadamard’s inequality gives $\det R \leq \prod_i R_{ii} = 1$ for a positive semi-definite matrix. A more telling proof goes through the spectrum: $\det R = \prod_k \lambda_k$ with $\sum_k \lambda_k = \text{tr } R = N$, and the arithmetic–geometric mean inequality gives $\prod_k \lambda_k \leq (\sum_k \lambda_k / N)^N = 1$, with equality if and only if all eigenvalues equal 1, that is $R = I$. Hence $H^{dep} \leq 0$: *correlation always destroys entropy*.

In the bivariate case, $\det R = 1 - \rho^2$ and $H^{dep} = \frac{1}{2} \log(1 - \rho^2)$; for $\rho = 0.9$ this is -0.83 nat. Note that $-H^{dep}$ is then exactly the mutual information between the two assets: correlation is shared information, hence destroyed entropy.

Reading note on the tables. The H^{cov} columns of Table 4 and following report $H^{vol} + H^{dep}$ and **omit the constant** $\frac{N}{2} \log(2\pi e)$, which equals 28.38 nats for $N = 20$. This constant offset is immaterial since only differences between regimes and changes over time are interpreted, but it explains why the reported levels are negative.

C The entropy deficit: why $\mathcal{D} = h(q) - h(p)$

The Kullback–Leibler divergence between a law p and a reference law q is $D_{KL}(p \parallel q) = \int p \log(p/q)$, positive and zero if and only if $p = q$ almost everywhere. It is neither symmetric nor equipped with a triangle inequality: it is not a distance, hence the term divergence. The letters “KL” are the initials of Kullback and Leibler.

Proposition 4. *If $q = \mathcal{N}(\mu, \Sigma)$ has the first two moments of p , then $D_{KL}(p \parallel q) = h(q) - h(p)$.*

Proof. $D_{KL}(p \parallel q) = -h(p) - \mathbb{E}_p[\log q(X)]$. Now $\log q$ is exactly quadratic in x , so its expectation depends only on the mean and covariance of the law under which it is evaluated. Since p and q share these two moments, $\mathbb{E}_p[(X - \mu)' \Sigma^{-1} (X - \mu)] = \text{tr}(\Sigma^{-1} \Sigma) = N = \mathbb{E}_q[\cdot]$, hence $\mathbb{E}_p[\log q] = \mathbb{E}_q[\log q] = -h(q)$ and therefore $D_{KL}(p \parallel q) = h(q) - h(p)$. $\square$

The moment-matching assumption is essential: without it the identity fails. This is the whole content of Proposition 1, read as a measurement rather than as an existence theorem. Economically, $\mathcal{D}$ is the price in nats of the lie “the market is Gaussian”: the apparent uncertainty that the Gaussian description overstates, because the true law is more concentrated than its covariance alone suggests.

D The structural index is a divergence

Proposition 5. *With $q_i = \mathcal{N}(\mu_i, \sigma_i^2)$ the Gaussian marginals with the same scales as p ,*

$$D_{KL}(p \parallel \prod_i q_i) = \mathcal{D} - H^{dep} = \mathcal{J}.$$

Proof. By definition,

$$\begin{aligned} D_{KL}(p \parallel \prod_i q_i) &= -h(p) - \mathbb{E}_p[\sum_i \log q_i(X_i)] = -h(p) + \sum_i \left[\frac{1}{2} \log(2\pi\sigma_i^2) + \frac{1}{2} \right] \\ &= -h(p) + \sum_i \log \sigma_i + \frac{N}{2} \log(2\pi e), \end{aligned}$$

using $\mathbb{E}_p[(X_i - \mu_i)^2] = \sigma_i^2$, true by construction. Substituting equation (6), i.e. $h(p) = \frac{N}{2} \log(2\pi e) + \sum_i \log \sigma_i + H^{dep} - \mathcal{D}$, the scale terms and the constant telescope and $\mathcal{D} - H^{dep}$ remains. $\square$

This is an exact identity, not an approximation, and it is what justifies Definition 1. Removing the scale channel from (6) might look like a fudge; the object that remains turns out to be a Kullback–Leibler divergence in good standing, and therefore inherits three properties for free: it is positive; it is zero if and only if returns are independent *and* Gaussian, which provides a well-defined rather than arbitrary reference point; and it is invariant under asset-by-asset changes of units, since p and $\prod_i q_i$ transform in the same way.

It is useful to set this identity against the general chain rule $D_{KL}(p \parallel \prod_i q_i) = \sum_i D_{KL}(p_i \parallel q_i) + I(x)$, where $I(x) = D_{KL}(p \parallel \prod_i p_i)$ is the multi-information. The first term is the sum of marginal deficits $\sum_i \mathcal{D}_1(x_i)$, the second the total dependence. The estimator used in the paper estimates the first term with the approximation of Hyvärinen (1998) and replaces the second by its linear part $-H^{dep}$; nonlinear tail dependence is therefore omitted from $\mathcal{J}$ and measured separately through a co-exceedance rate.

E Mutual information: the two propositions and an example

Proof of Proposition 2. By definition of conditional entropy, $\mathbb{E}_Y[h(X | Y)] = h(X, Y) - h(Y)$, and by (9), $I(X; Y) = h(X) + h(Y) - h(X, Y)$. Adding the two, $\mathbb{E}_Y[h(X | Y)] = h(X) - I(X; Y)$. Positivity of I follows from that of the Kullback–Leibler divergence, of which it is a special case, with equality if and only if $p_{XY} = p_X \otimes p_Y$.

Proof of Proposition 3. Immediate: an additional constraint can only shrink the feasible set, and the supremum of a functional over a smaller set cannot increase.

Why the two are not equivalent. The first quantifies but holds only on average; the second holds realisation by realisation but concerns the ceiling and quantifies nothing. A discrete counterexample suffices to show that a particular realisation can *increase* entropy. Let X be binary with $\mathbb{P}(X = a) = 0.99$, with entropy $H(X) = 0.056$ nat. Let a signal Y , with probability 0.98, remove all doubt ($H(X | Y = 0) = 0$), and with probability 0.02 lead to $\mathbb{P}(X = a | Y = 1) = 1/2$, i.e. $H(X | Y = 1) = \log 2 = 0.693$ nat — twelve times the prior entropy. The average is nevertheless $0.98 \times 0 + 0.02 \times 0.693 = 0.014$ nat, well below 0.056, and $I(X; Y) = 0.042$ nat. This is exactly why the paper predicts a jump component that is *mainly*, not exclusively, one-sided.

Worked Gaussian example. Let $X \sim \mathcal{N}(0, \sigma^2)$ be tomorrow’s return and $Y = X + \varepsilon$ a noisy signal, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ independent. Then

$$I(X; Y) = \frac{1}{2} \log \left(1 + \frac{\sigma^2}{\sigma_\varepsilon^2} \right), \quad \text{Var}(X | Y) = \frac{\sigma^2 \sigma_\varepsilon^2}{\sigma^2 + \sigma_\varepsilon^2}, \quad (17)$$

the posterior variance not depending on the realisation y . With $\sigma = 1\%$ per day:

Signal quality	σ_ε	$I(X; Y)$ (nats)	posterior vol.	$\sqrt{2\eta}$ equivalent
sharp	0.5%	0.805	0.45%	1.27σ
medium	1.0%	0.347	0.71%	0.83σ
vague	3.0%	0.053	0.95%	0.32σ

A factor of fifteen separates the size of the entropy jump caused by decisive news from that caused by a rumour. The last column translates each jump onto the severity scale of Section 8: it is the directional move that a scenario of the same radius would produce.

F Exploratory: composition of crises and of scenarios

This appendix gathers a result that clears only one of the two nulls and the use made of it to compose a stress scenario. We separate it from the body of the paper because nothing above depends on it, and because it does not meet the standard the paper sets itself everywhere else. It should be read as a lead, not as a result.

F.1 H8: the composition of the crisis budget

The seven hypotheses above treat each channel on average over a whole regime. The question here is different: *inside a given crisis*, which channel does the work? A jump detected in $\mathcal{J}$ is rarely an isolated event — it is part of a run of nearby jumps, as Figure 3 already shows — so we first merge jumps less than 40 trading days apart into *episodes*, then decompose each episode’s move, from onset to peak, over the three channels:

$$\Delta\mathcal{J}_{\text{episode}} = \underbrace{\Delta(-H^{\text{dep}})}_{\text{dependence}} + \underbrace{\Delta d_{\text{odd}}}_{\text{skewness}} + \underbrace{\Delta d_{\text{even}}}_{\text{tail}}, \quad (18)$$

an exact identity from Definition 1 and the odd/even decomposition of negentropy behind (7) (`episode_channel_budget`, `src/jumps.py`): the three shares sum to one by construction, so this is not an approximation of a per-crisis budget but the budget itself.

The raw fact. On all three panels, the *skewness* channel is almost never the dominant contributor (1 episode in 139, all panels combined) — one more piece of evidence against P2 as the dominant mechanism, on top of the already marginal result of H5. The split between *dependence* and *tail*, on the other hand, looks like a real typology: on the main panel, dependence dominates 46% of the 65 detected episodes; on Kenneth French’s 49-industry panel, 79% of 61 episodes; on `EuStockMarkets`, 69% of 13 episodes.

Panel (episodes)	Observed	i.i.d. null mean	5–95 %	Verdict
S&P 500 (65)	0.46	0.26	0.18–0.33	clears
FF49 (61)	0.79	0.43	0.33–0.53	clears
<code>EuStock</code> (13)	0.69	0.57	0.34–0.77	does not clear

Table 10: H8, i.i.d. null: share of detected episodes whose dominant contributor is the dependence channel. Two hundred replications (S&P 500, FF49), fifteen (`EuStock`). `scripts/episode_null.py`.

Table 10 confirms the fact on all three panels: the observed value exceeds all (S&P 500, FF49) or most (`EuStock`) of the i.i.d. replications. `EuStockMarkets` does not formally clear the null — the observed value lies inside its 90% band — but the direction is right, and the panel has thirteen episodes against sixty-odd for the other two: the same low-power caveat made elsewhere in this paper for this panel. Two panels that clear clearly and a third consistent in direction: H8 is retained, *in the sense of the i.i.d. null*.

Why only the i.i.d. null. An additional decomposition was tested on the main panel: what fraction of episodes does the 21-day block null — the one that preserves short-term volatility clustering — also reproduce as dependence-dominated? It does reproduce it: the observed value (0.46) this time falls inside the band of the block null (0.43, 0.35–0.57). Table 11 shows that this is not

an artefact of the arbitrary choice of 21 days: clearance switches from clear (block 5) to marginal (block 10) to clear failure (block 21), and the null *even exceeds* the observed value at blocks 42 and 63, before easing slightly at block 126 — a sign that blocks this long start to replay whole historical episodes rather than destroy their structure. Ordinary volatility clustering alone therefore plausibly suffices to produce the dependence/tail typology, whatever block length from 21 days is tested — the same verdict as already reached for H3 and H6.

Block length (days)	null mean	5–95 %	$p(\text{obs.} \geq \text{null})$	Verdict
1 (i.i.d.)	0.26	0.18–0.33	0.00	clears
5	0.31	0.23–0.41	0.00	clears
10	0.35	0.27–0.45	0.08	marginal
21	0.43	0.33–0.54	0.33	fails
42	0.49	0.40–0.60	0.67	fails, null above
63	0.51	0.43–0.56	0.75	fails, null above
126	0.47	0.40–0.56	0.50	fails

Table 11: Sensitivity to block length, S&P 500: observed share of dependence-dominated episodes (0.46, constant across the row) against the block null for seven lengths. Two hundred replications for blocks 1 and 21, shared with Table 10, twelve for the five intermediate and long lengths swept specifically here.

This finding could have closed the lead — it is the same verdict as H3 and H6 — but the stress test of Section 8 was never judged against a null: it is a measurement convention plus a convex optimisation (end of Section 7.1), and nobody claims it describes a dynamics. What H8 brings to that convention is therefore not “dependence causes crises” — a claim the block null forbids — but a more modest observation, sufficient to calibrate a composition: *the budget of entropy destruction observed on this sample is not the one a resampling without any temporal structure would produce*. Section F.2 uses this fact, and only this fact, with the scope explicitly narrowed to the i.i.d. null — a deliberate decision of scope, not a selection of the nulls that suit the result.

Severity does not predict composition. One last natural question: does the dependence/tail split drift with episode size, so that one could calibrate a composition by severity rather than an average composition? Table 12 splits the 65 episodes of the main panel into quintiles of $\Delta\mathcal{J}$ and averages the dependence share in each. The pattern is not monotone — the dip in the second quintile is real, not a rounding artefact — and the correlation between episode size and dependence share is indistinguishable from zero (Spearman 0.05, Pearson 0.03). There is no “bigger crisis, more dependence” law to extract from this sample. The calibrated composition of Section F.2 therefore uses the average regime split (H1) as anchor, and the split measured on the more severe half of episodes (this table) as the only defensible refinement.

F.2 Calibrated composition: which channel pays for severity

Construction (13) constrains only the projection of the portfolio onto the KL ball — it fixes the stressed portfolio mean and volatility, m and s — and is silent on how the underlying covariance should deform, asset by asset, to reach them. The simplest choice compatible with that constraint, the one that produces Tables 8 and 9, is a uniform dilation: every marginal volatility multiplied by the same factor v , correlations unchanged. That is a construction default, not a statement about how crises behave. H1 says how they behave: the dependence channel moves on its own account, and it is the only regime channel in the whole paper to clear both nulls (Table 4). H8 refines that

Quintile of $\Delta\mathcal{J}$ (S&P 500)	dependence share	skewness share	tail share
q1 (calmest)	0.53	0.10	0.37
q2	0.17	0.02	0.80
q3	0.37	0.04	0.59
q4	0.50	0.02	0.48
q5 (most severe)	0.49	0.02	0.49

Table 12: Average share of each channel by quintile of episode severity, 13 episodes per quintile, main panel.

fact at the episode scale, with the narrowed scope its own null demands (Section F.1). This section moves these two facts from the “result” column to the “calibration” column, and relies exclusively on the two hypotheses their respective nulls retain.

The mixture stays exact.

Proposition 6 (One severity, several compositions). *Let $\Sigma_a, \Sigma_b \succeq 0$ be two covariances and w a vector of portfolio weights such that $w^\top \Sigma_a w = w^\top \Sigma_b w = s^2$. Then for any $\alpha \in [0, 1]$, $\Sigma_\alpha \equiv (1 - \alpha)\Sigma_a + \alpha\Sigma_b$ still satisfies $w^\top \Sigma_\alpha w = s^2$.*

Proof. $w^\top \Sigma w$ is linear in Σ , so $w^\top \Sigma_\alpha w = (1 - \alpha)w^\top \Sigma_a w + \alpha w^\top \Sigma_b w = (1 - \alpha)s^2 + \alpha s^2 = s^2$. $\square$

An immediate and usable consequence: one can vary *how* the severity s is distributed across assets without ever changing s . The mixture parameter, $\alpha \in [0, 1]$, is what we call the *dependence share*.

A physical ceiling. Correlation alone cannot carry any severity. At fixed marginal volatilities σ_i , $w^\top \Sigma w = \sum_i w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j \sigma_i \sigma_j \rho_{ij}$ is maximised, over valid correlation matrices, by $\rho_{ij} = 1$ for all $i \neq j$ — perfect dependence — where it equals $(\sum_i w_i \sigma_i)^2$. On the main panel, this ceiling is only about 1.5 times today’s volatility: a diversified portfolio of twenty stocks simply cannot reach high severity through correlation alone, however “dependent” the underlying crisis — which Table 8 shows on this panel: the 1.37 multiplier of the 5-year rung is still reachable through correlation alone, the 1.63 of the 10-year rung is not (implied average correlation of 1.1). The construction retained accounts for this: correlation is pushed towards its ceiling ($\rho = 1$, capped) in proportion to the requested dependence share, and the remaining gap, unavoidable beyond that ceiling, is filled by a final uniform dilation, so that Proposition 6 applies whatever the requested share (`calibrated_covariance`, `src/stress.py`).

Where the dependence share comes from. Two sources, neither arbitrary.

- **H1 anchor.** The regime share of Table 4: $|-0.99|/(|-0.99| + |5.97|) = 0.14$. It is the most robust fact in the whole paper — it clears both nulls — turned into a calibration parameter instead of remaining a result that is quoted and then dropped.
- **H8 refinement.** The same share, measured only on the more severe half of detected episodes: 0.49, more than three times the regime average — consistent with Table 12, where quintiles 4 and 5 clearly dominate quintile 2, without claiming a monotone law.

Composition (10-year scenario, $\eta = 0.41$)	dependence share	ambient price (nats)
default (KL ball, pure dilation)	0.00	6.80
H1-calibrated (regime average)	0.14	2.88
H8-refined (severe episodes)	0.49	0.71

Table 13: Ambient price, in nats, of three compositions delivering the same portfolio severity (mean, volatility, $ES_{99} = 5.9\%$, identical across the three rows by construction). `scripts/calibrated_stress_demo.py`, S&P 500 panel. On French’s industries ($\eta = 0.37$): 14.7, 8.3 and 5.8 nats for shares of 0.00, 0.20 and 0.75.

What this changes, at fixed severity. Table 13 prices the 10-year scenario of Table 8 under these compositions. The price in *ambient* nats — the full KL divergence between the stressed scenario and today’s distribution, computable in full dimension, not to be confused with the radius η which constrains only the projection — *decreases* as the dependence share increases, rather than increasing. This is not a computational anomaly: uniform dilation is the cheapest choice only in the sense of the projected constraint; it is not a proven minimum of the ambient divergence. That minimum exists — the first-order conditions of the problem “minimise $\text{tr}(\Sigma_1 \Sigma_0^{-1}) - \log \det \Sigma_1$ subject to $w^\top \Sigma_1 w = s^2$ ” give a rank-one perturbation of the *precision* matrix, $\Sigma_1^{-1} = \Sigma_0^{-1} - \lambda w w^\top$ — but that object is not interpretable asset by asset and is not used here. The empirical finding is therefore this: on this panel, moving towards the historically observed composition also moves, without this being sought, towards that unused minimum. A risk manager who assumes uniform dilation by default is not protecting themselves by being prudent — the composition anchored on H1 and refined by H8 is both more realistic *and* cheaper on the market’s information scale.

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